



DEPARTMENT OF COMPUTER SCIENCE
MSc in DATA SCIENCE AND BUSINESS INFORMATICS

***Physical and Behavioral Correlates of Problematic Internet Use: An Explainable Machine
Learning and Time Series Study on Children and Adolescents***

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0. Abstract

In this paper our goal is to forecast the Severity Impairment Index (sii) reflecting problematic internet usage by children and adolescents aged 5–22 years old; the total number of subjects included in our dataset is of 8,460 individuals. The features we will use to forecast sii span across the subject's physical health, their level of fitness, body composition, and sleep disturbances, with the specific intention of avoiding the use of PCIAT score which would cause data leakage. For this purpose, a regression task has been performed (8 models tuned with Optuna, stacking ensemble), a quadratic weighted kappa of .3712 has been achieved. In the classification task (8 models, Macro F1 optimised), a maximum Macro F1 = .38 and macro auroc = .71 have been achieved via XGBoost. All four of the tested explainability methods (SHAP, LIME, TREPAN, LORE) converged on sleep disturbances and self reported daily screen time as the top predictors of high sii. The main limitation of this study was making correct predictions about class 3 ("severe", n=70), as it had very few subjects.

1. Introduction

1.1 Task Definition and Dataset Overview.

The Child Mind Institute (CMI) problematic internet use competition was designed to determine if data deriving from children's physical health can be used to forecast the degree of problematic internet usage. The response variable in this dataset is the Severity Impairment Index (sii). Sii is an ordinal feature scale (0=none, 1=mild, 2=moderate, 3=severe) derived from the Parent-Child Internet Addiction Test (PCIAT).

In other words, once the PCIAT item scores are summed together to derive the PCIAT total score, this total will uniquely determine the class label. This constitutes an information leakage hazard. However, all pipeline(s) have been carefully constructed to mitigate this issue.

Study design. Two separate predictive tasks have been developed:

- **Regression:** treat sii as a continuous ordinal variable and round the predicted values to an integer label; evaluate using Quadratic Weighted Kappa (QWK).
- **Classification:** predict sii as a four-class nominal target; evaluate using Macro F1, Accuracy, and Macro AUC.

Although the two predictive tasks utilize different evaluation metrics, both tasks are use the same raw feature set, the same 80/20 stratified train-test split, and the same outlier detection logic. However, each task uses a different method for handling missing data, a different approach to dealing with class balance issues, and a different family of machine learning models.

PCIAT leakage: important note. There is a trivial model that simply keeps any of the PCIAT columns in the predictor feature space and that achieves a nearly perfect QWK score ($\approx .99$), since sii is calculated directly from those columns. All models reported in this paper do not make use of such feature: indeed, PCIAT-PCIAT_Total as well as all 20 individual PCIAT item columns have been deleted prior to performing train-test splits or any other operations.

Dataset. The tabular data set (cmi_internet.csv) was created by modifying the original Kaggle CMI competition data set to include artificial rows. In total, this modified dataset contained **8,460 rows** and **82 columns** (in addition to those relating to the subject's id and the PCIAT block).

Feature overview. The predictive features included within the study span seven domains:

Family	Prefix	Content
Demographics	Basic_Demos-	Age (years), Sex (binary)
Physical measurements	Physical-	Height, Weight, BMI, Waist circumference, Heart rate, Systolic/Diastolic BP
Fitness endurance	Fitness_Endurance-	Max stage reached, Time in minutes and seconds (20-metre shuttle run)
Fitness grip / flexibility	FGC-	Curl-ups (CU), grip strength dominance/non-dominance (GSD/GSND), push-ups (PU), sit-reach left/right (SRL/SRR), trunk lift (TL), each paired with a fitness zone score
Bioelectrical impedance	BIA-	BMI, BMR, DEE, ECW/ICW, FFM, FFMI, FMI, Fat, LDM, LST, SMM, TBW, activity level, body frame
Physical activity questionnaire	PAQ_A-, PAQ_C-	Self-reported physical activity totals (adolescent and child forms)
Sleep disturbance	SDS-	SDS Total Raw score and T-score
Internet use	PreInt_EduHx-	Self-reported daily computer/internet hours
Target / leakage	PCIAT-, sii	PCIAT item scores (dropped), PCIAT Total (dropped), sii (target)

Target distribution. Of the total number of rows ($n = 8,460$), none had a null value *sii*. Although each row includes a target classification, some predictor variables may be missing. Therefore, the target distribution is heavily skewed toward Classes 0, with a solid but inferior presence of class 1 and 2, and with Class 3 being extremely rare ($\approx 1\%$). As such, this imbalance presents an ongoing issue when attempting to improve recall performance in identifying the medical most relevant category.

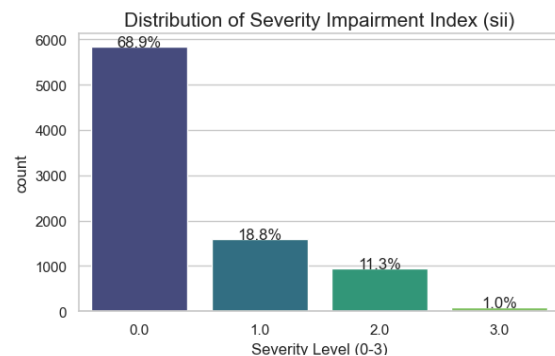
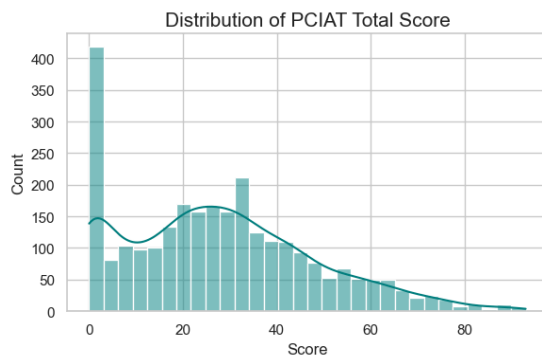
Class	Label	Train count	Train %
0	None	4,666	68.9%
1	Mild	1,270	18.8%
2	Moderate	762	11.3%
3	Severe	70	1.0%

Audit of data entries. Seven features with known physical bounds are range-checked before any modelling step: values outside these limits are assigned as NaN before splitting. Thus, all entries falling out of the ranges specified, which are very likely to be incorrect, are treated as missing values. After conducting the audit, the dataset continues to consist of **8,460 rows $\times$ 82 columns**.

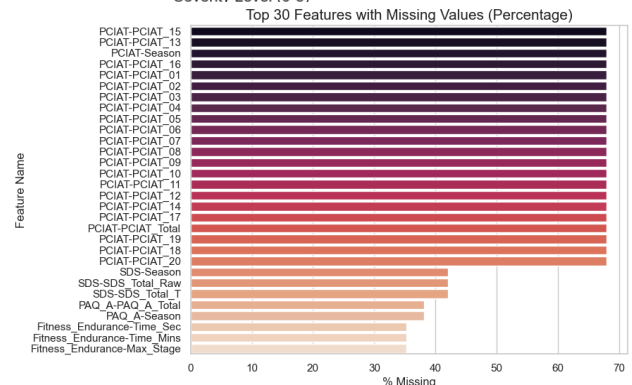
Feature	Valid range
Basic_Demos-Age	[5, 22]
BIA-BIA_BMI	[10, 50]
Physical-HeartRate	[30, 220]
Physical-Systolic_BP	[50, 200]
Physical-Diastolic_BP	[30, 130]
Physical-Height	[30, ∞)
Physical-Weight	[10, ∞)

2. Exploratory Data Analysis

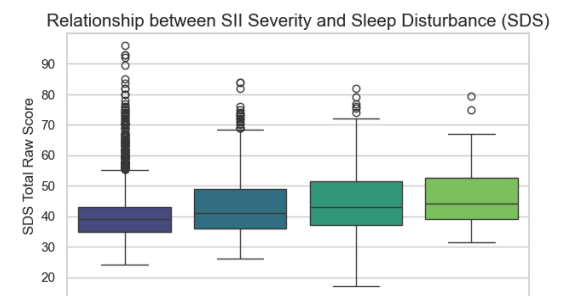
The *sii* labels follow a skewed multinomial distribution located at the low-end of the severity spectrum. The total score of the PCIAT shows a right-skew similar to other distributions of a population exhibiting minimal to no internet problematic usage.



Missingness Values. There are missing values throughout the entire dataset, and the relative percentage varies by feature. For example, many of features related to fitness testing (i.e. FGC-, Fitness_Endurance-) display a high level missing values. Additionally, many of the individual PCIAT items were either completely or partially missing. Other features with missing values include the sleep disturbances score (SDS-SDS_Total_Raw) and bioimpedance battery (BIA-). However, no single category exceeds a 50% rate of missing values in the training set after leakage coming from PCIAT was removed. Therefore, no columns were removed due to missing values, but two BIA- features that were removed through the Spearman correlation filter.

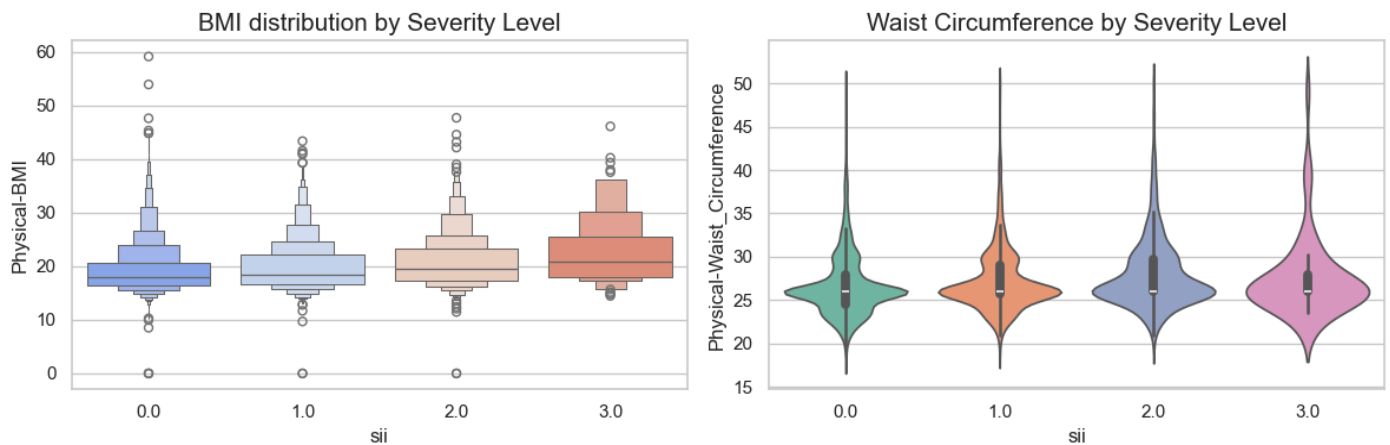


Relationships with the target: *sii* and SDS. A relation between the Sleep Disturbance Score (SDS) and *sii*. For example, the median SDS score for Class 3 is significantly greater than that of Class 0. Also, the interquartile ranges clearly separate each of the different classes. We can therefore affirm that

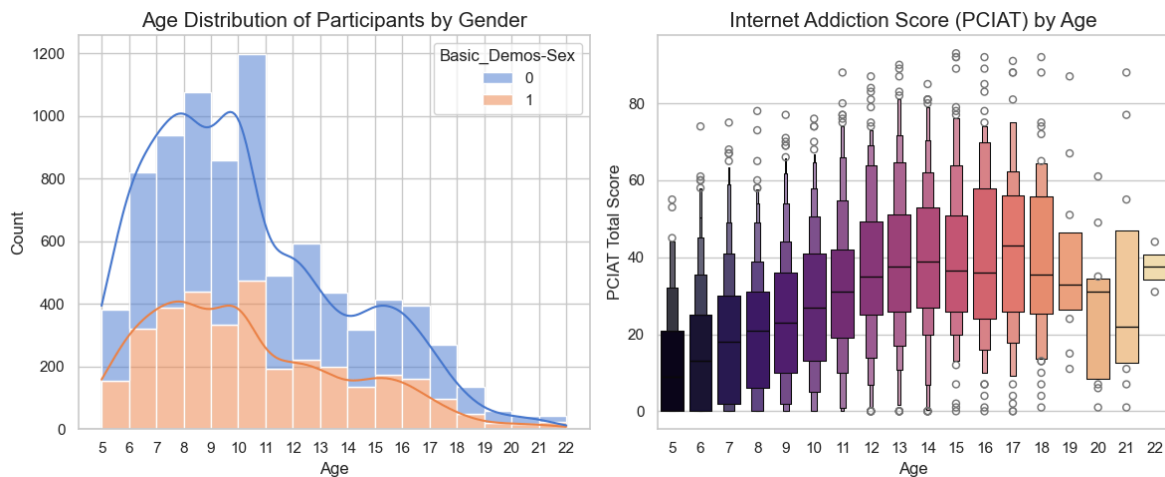


children who engage in problematic use of the internet tend to also display disruptions in their sleep patterns.

Relationships with the target: sii and physical fitness. There is a generally weak, though consistent inverse relationship between physical fitness and sii. Specifically, individuals with higher sii tend to also have higher BMIs (Physical-BMI), larger waists (Physical-Waist_Circumference), and lower fitness zone scores (e.g., FGC-FGC_CU_Zone, indicating cardiovascular fitness) than those with less severe levels of sii. These findings can support the hypothesis that sedentary screen time displaces physical activity.



Relationships with the target: sii across different age and sex. Sex do not strongly drive PCIAT Total scores but age does; scores go up over years which suggests two scenarios: one is that damage accumulates and the other which seems more likely is adoption of different electronic devices (smartphones, tablets, consoles and computers etc.) on which kids and teens spend their time shifting from outdoor play to online play and from going out, to spending time on social media. Something to consider is that data after age 18 is less reliable because smaller sample sizes, this make interpretation of consequences for adults unreliable and highly volatile.



Correlation structure. A too wide correlation matrix covering all numeric features shows mainly that::

- High internal correlation amongst all BIA features (for example, BIA-BIA_FFM and BIA-BIA_LST both showed Spearman $|\rho| > 0.90$ as demonstrated in the pre-processing stage);
- Moderate positive Spearman correlations between SDS-SDS_Total_Raw / SDS-SDS_Total_T and sii;
- Moderate negative Spearman correlations between fitness zone scores (FGC-) and sii;
- Mild positive correlation exists between PreInt_EduHx-computerinternet_hoursday and sii, indicating that self-reported hours spent daily engaging in screens are positively correlated with sii.

3. Preprocessing and Outlier Detection

3.1 Leakage Removal and Feature Selection

Schema-level drops (before split). Three types of columns were dropped prior to any train-test split:

1. PCIAT individual items: 20 columns (PCIAT-PCIAT_01 ... PCIAT-PCIAT_20) were dropped because their sum directly determines sii;
2. Season columns: dropped as they had no added predictive value;
3. CGAS proxy: CGAS-CGAS_Score was dropped to again avoid leakage.

Additionally, PCIAT-PCIAT_Total and the Administrative ID column were removed from the feature matrix. Once done, the numeric column size went from 82 down to 48 (including sii).

Train/test split. A stratified 80-20 split on sii is applied with random_state=42, yielding:

- **Training pool:** 6,768 rows $\times$ 47 features;
- **Test set:** 1,692 rows $\times$ 47 features.

Train-only data-dependent filters (after the split, fitted on train data only and then applied to test):

1. **Missing values >50% filter:** No feature had more than 50% of missing values, hence all 47 features have been kept;
2. **Spearman $|\rho| > 0.90$ filter:** BIA-BIA_FFM and BIA-BIA_LST are dropped (both measure lean tissue mass and are collinear with other BIA totals). The final feature dimensionality is **45**.

3.2 Imputation Tuning

Missing values are imputed via a tree-based iterative strategy. An **Optuna** study of 100 trials evaluates two imputer families:

- **Extra Trees Imputer:** IterativeImputer with backing provided by ExtraTreesRegressor, with tuning of max_iter $\in [3, 10]$, initial_strategy $\in \{\text{mean, median}\}$, n_estimators $\in [5, 20]$ and max_depth $\in [2, 5]$.
- **MICE Imputer:** IterativeImputer with backing by ExtraTreesRegressor, with tuning of max_iter $\in [5, 15]$ and similar trees parameters.

Tuning is scored on a masked-reconstruction objective: 10% of observed values are hidden and the imputer's ability to reconstruct them (NRMSE) is evaluated. The study was run once over the entire training pool. Then the optimal choice of imputer type and parameters is used to refit within each CV Fold.

Regression best (from Optuna):

{'imputer_choice': 'extra_trees', 'et_max_iter': 6, 'et_initial_strategy': 'median', 'et_estimators': 20, 'et_max_depth': 5. CV QWK Score: 0.3730.

Active imputer in regression pipeline (03):

Stability analysis indicated that the MICE configuration resulted in more constant residuals throughout folds. Therefore the active configuration is:

{'imputer_choice': 'mice', 'mice_max_iter': 10, 'mice_initial_strategy': 'median', 'mice_et_estimators': 20, 'mice_et_max_depth': 4}

Classification best (from Optuna):

{'imputer_choice': 'mice', 'mice_max_iter': 6, 'mice_initial_strategy': 'median', 'mice_et_estimators': 19, 'mice_et_max_depth': 5}

Caveat. The study of the optimal imputer is conducted upon the complete training pool, which means it sees rows that will later be part of CV validation folds. Because the imputer is unsupervised (the imputer never sees y), the practical bias is small but to consider slightly optimistic.

3.3 Consensus Outlier Detection

Method. Ten outlier detectors coming from different families were run in parallel over the training pool. Each detector scored every row of the training set. We consider an observation as an outlier if its score was greater than the **99th percentile** (top 1%) of the distribution of that method.

#	Method	Family
1	HBOS	Naive/Statistical
2	Grubbs proxy (Mahalanobis via PCA)	Statistical
3	Arning SF (squared distance from mean)	Deviation
4	ISODEPTH (Tukey depth via 2D PCA)	Depth-based
5	kNN (largest distance to k neighbours)	Distance-based
6	LOF	Density-based
7	CBLOF	Clustering-based
8	ABOD	Angle-based / High-dimensional
9	LODA	Ensemble
10	IForest	Model-based

A consensus vote requires ≥ 9 out of 10 detectors to flag a row together. This high threshold avoids an individual methods' bias to drive the overall choice about removals.

Regression outlier results.

CONSENSUS OUTLIER DETECTION (10 METHODS)

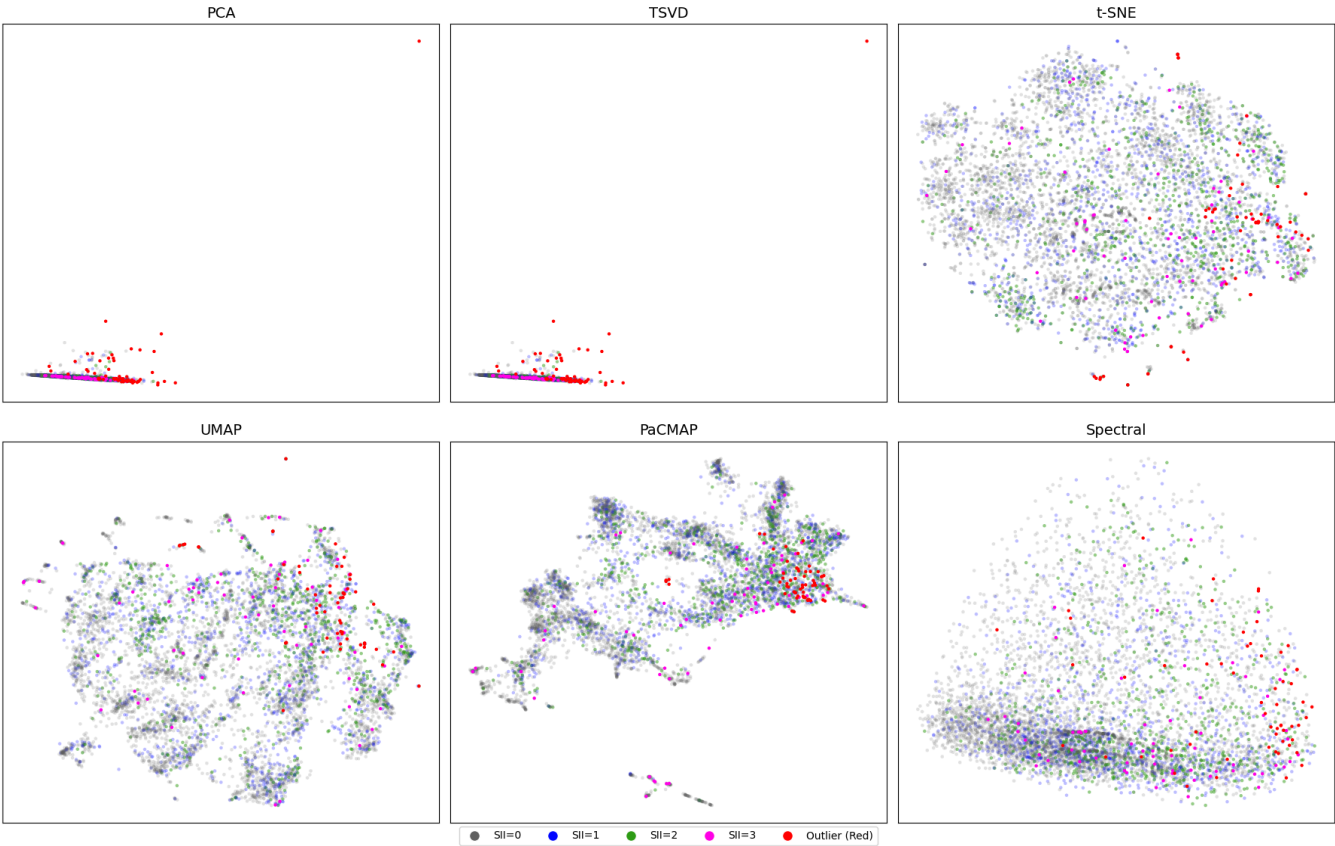
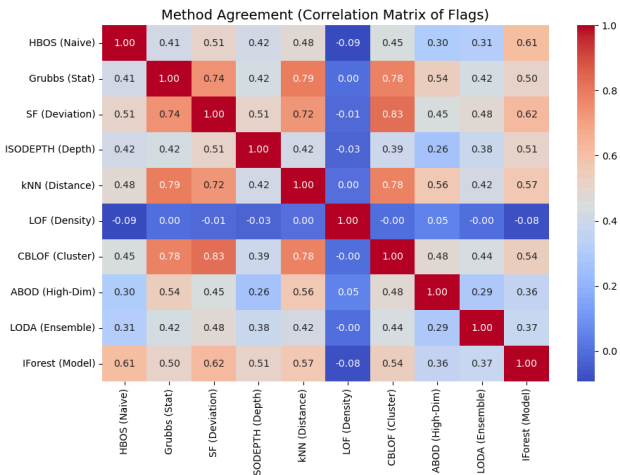
Total Outliers: 59/6768 (0.87%)

Class 0: 33 outliers (0.71% of class, 0.49% of total)

Class 1: 8 outliers (0.63% of class, 0.12% of total)

Class 2: 17 outliers (2.23% of class, 0.25% of total)

Class 3: 1 outlier (1.43% of class, 0.01% of total)



Classification outlier results:

10-TYPEOLOGY CONSENSUS OUTLIER DETECTION

Consensus threshold: $\geq 9 / 10$ agreement (top 1% per method)

Total Outliers: 54 / 6768 (0.80%)

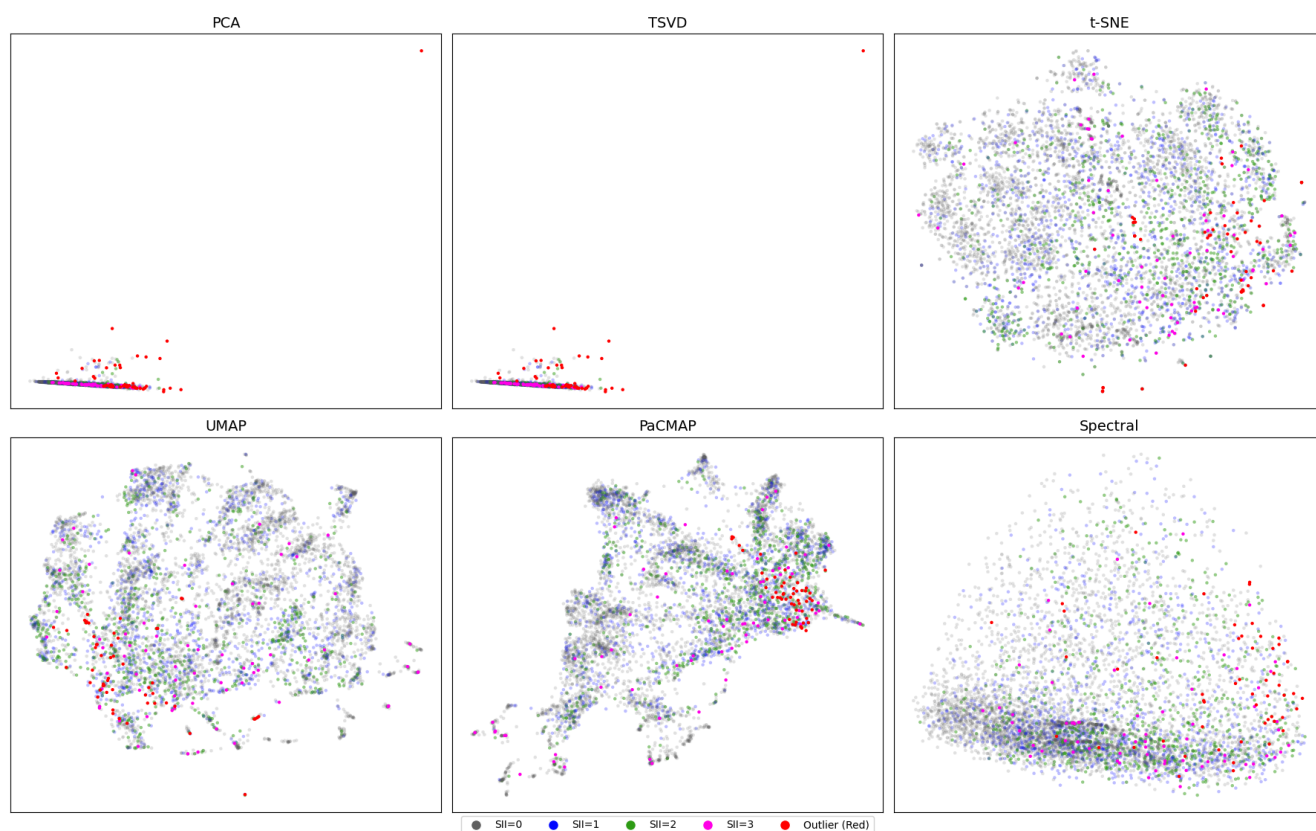
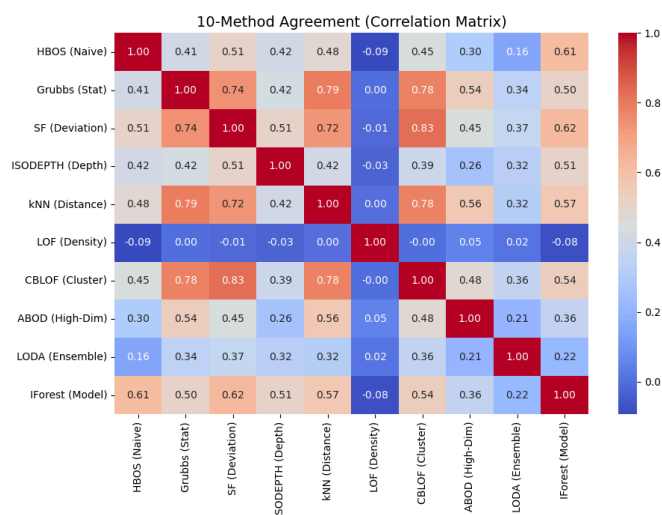
Class 0: 29 outliers (0.62% of class)

Class 1: 11 outliers (0.87% of class)

Class 2: 13 outliers (1.71% of class)

Class 3: 1 outlier (1.43% of class)

The small difference between regression and classification count (59 vs. 54) is due to each pipeline running the detector independently on it's own imputed representation. Both pipelines store outlier flags into the exported CSV (is_outlier = 1). Models trained in notebooks 03/04/05/06 will drop flagged rows from their training folds.



Dimensionality reduction visualisation. Outliers have been viewed in six two-dimensional representations by PCA, Truncated SVD, t-SNE, UMAP, PaCMAP and Spectral Embedding. Outlier points (red) did not group themselves within one area, validating that the consensus methodology identified global outliers instead of a single area of the feature space.

3.4 Imbalanced Learning

A lightweight proxy classifier (LGBMClassifier with 100 trees, max_depth=5) embedded in an **imblearn Pipeline** was assessed using 5-fold Stratified Cross-validation. Seven resampling setups were compared.

Oversampling methods. ADASYN achieved the highest Macro F1 (0.3261) amongst the sampled methods. It generated synthetic minority samples in the vicinity of the decision boundary where the classifier was weakest. SMOTE followed close behind. They both improved significantly relative to the baseline (0.2844) indicating that synthetic oversampling helped the proxy model find minority-class patterns.

Method	Type	Macro F1	Macro Recall	Macro Precision
Baseline (No Resampling)	-	0.2844	0.2890	0.3460
SMOTE	Oversampling	0.3078	0.3040	0.3489
ADASYN	Oversampling	0.3261	0.3182	0.3713
ENN	Undersampling	0.2208	0.2751	0.1862
Tomek Links	Undersampling	0.2815	0.2873	0.3541
SMOTE + ENN	Combined	0.3221	0.3618	0.3448
SMOTE + Tomek	Combined	0.3151	0.3103	0.3508

Oversampling methods. ADASYN achieved the highest Macro F1 (0.3261) amongst the sampled methods. It generated synthetic minority samples in the vicinity of the decision boundary where the classifier was weakest. SMOTE followed close behind. They both improved significantly relative to the baseline (0.2844) indicating that synthetic oversampling helped the proxy model find minority-class patterns.

What ADASYN's advantage reveals about data geometry. That ADASYN out performed uniform SMOTE is itself a diagnostic result regarding the dataset. SMOTE generates synthetic minority samples equally spaced along line segments between existing minority samples; ADASYN generates synthetic minority samples on minority samples that are hardest to classify, which are those bounded by majority-class neighbors. The better performance of ADASYN is thus evidence that minority-class samples ($s_{ii} \geq 1$) in this feature space do not cluster as compactly into a separable area, but they scatter towards the majority-class boundary.

Undersampling. ENN produced a very poor outcome (Macro F1 = 0.2208), as it deletes majority-class samples whose labels conflict with their neighbors. However, since there are four classes with one significantly more numerous than others, removing majority-class samples collapses all performance. Tomek Links had little effect, it removed only borderline majority-class instances, it did not differ from baseline at all.

Combined methods. The combination of SMOTE + ENN delivered good recall (0.36) while producing relatively good precision (0.34), so it was the best of all the combined techniques. SMOTE + Tomek was less effective than SMOTE alone.

Practical decision. Since the downstream classification models will utilize `class_weight='balanced'` (or `auto_class_weights='Balanced'` for Catboost), which is equivalent to inverse-frequency resampling at the sample weight level, explicit sampling does not occur in the last Optuna tuning pipeline. The imbalanced learning analysis was used to inform the decision to favor Macro F1 when evaluating the classification task, also because in various testing was clear that although oversampling improved recall, we noticed also an increase of false positive, for this reason `class_weight='balanced'` was the best trade off after extensive testing.

4. Regression

4.1 Pipeline Architecture

Each regression model was implemented into a preprocessing function called `preprocess_fold` which contains all of the steps taken within each cross-validation split. Within each split these steps include:

1. **Imputation:** tree based IterativeImputer using the MICE strategy;
2. **Standardisation:** StandardScaler fit to train dataset only;
3. **Optional PCA:** prior to training for linear or kernel based models (Ridge, SVR, Lasso, and Neural Networks) a PCA is performed for dimensionality reduction, and the number of dimensions is also defined in the Optuna search space;
4. **Sample weighting:** sample weights calculated using `compute_sample_weight('balanced', y)` are added to the model at time of training to address class imbalance issues;
5. **Threshold optimisation:** after training, the predictions of s_{ii} made on the train folds are rounded to 0, 1, 2, 3 by applying three threshold values optimized using a grid search ($\text{span} = \pm 0.5$, $\text{step} = 0.05$, $\text{iterations} = 3$) to maximize QWK on the training fold.

Cross validation utilizes a **stratified KFold** method with five splits (`StratifiedKFold(n_splits=5)`) using the integer value from the s_{ii} label for stratification. Validation will be done using QWK as the optimization metric.

4.2 Hyperparameter Tuning: Optuna Results

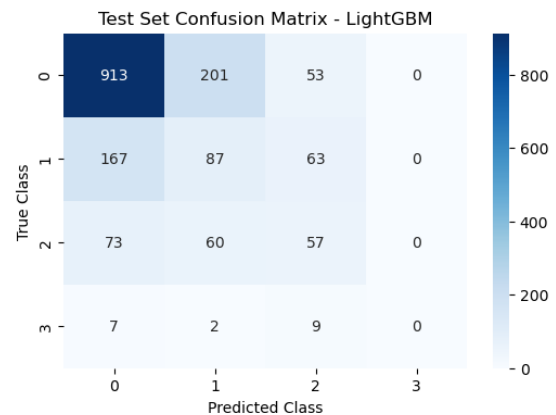
All regression models are tuned with Optuna (TPE sampler, maximise QWK). The search spaces and best CV QWK scores are:

Model	Trials	Key search space	Best CV QWK
LightGBM	50	learning_rate [1e-3, 0.1], num_leaves [20, 100], max_depth [3, 10], n_estimators [50, 300], reg_alpha/lambda log-uniform	0.3376
XGBoost	50	learning_rate [1e-3, 0.1], max_depth [3, 10], min_child_weight [1, 10], subsample [0.5, 1.0], n_estimators [50, 300]	0.2746
CatBoost	50	learning_rate [1e-3, 0.1], depth [4, 10], l2_leaf_reg log-uniform, iterations [50, 300]	0.2414
Random Forest	50	n_estimators [50, 300], max_depth [3, 20], min_samples_split [2, 20], min_samples_leaf [1, 20], max_features [0.1, 1.0]	0.3125
Ridge (+PCA)	50	alpha [1e-8, 10], pca_n_components [5, 45]	0.2692
SVR (+PCA)	20	C log-uniform, kernel $\in$ {linear, poly, rbf, sigmoid}, gamma $\in$ {scale, auto}, pca_n_components [5, 45]	0.2704
PyTorch MLP (+PCA)	20	n_layers [1, 3], units [32, 256], dropout [0, 0.5], lr log-uniform, epochs [50, 500], optimizer $\in$ {adamw, adam, sgd, rmsprop}	0.3098
Lasso (+PCA)	20	alpha [0.001, 10], pca_n_components [5, 45]	0.2675

4.3 Final Evaluation: Test Set QWK

After hundreds of trials in each Optuna study, for each model in the moment in which we evaluate on the holdout test set we take the best hyperparameter configuration that has maximized QWK in cross validation. Threshold optimization uses Nelder-Mead (scipy.optimize.minimize) on training-set. predictions.

Model	Test QWK
LightGBM	0.3613
Random Forest	0.3247
PyTorch MLP	0.3098
XGBoost	0.2946
Ridge	0.2885
SVR	0.2801
CatBoost	0.2714
Lasso	0.2675



Stacking ensemble. Uses LightGBM, XGBoost, CatBoost, Random Forest, Ridge and Lasso as base learners; Ridge is utilized as the meta learner of the predictions of all of the base learners. MLP has been omitted from the stack due to its suboptimal performance across CV metrics. A Nelder Mead algorithm (trial and error optimization method that stretch, shrink and shift the geometric shape made by the stacking guesses) is used to optimize the threshold values of the ensemble's continuous output.

Ensemble test QWK: 0.3712, a gain of +0.0099 QWK compared to the best single model (LightGBM), indicating that the errors produced by the base learners were complimentary.

Discussion. The regression QWK scores we obtained reflect how difficult it is to predict such an outcome. When removing PCIAT, there is little to no signal for a psychobehavioral outcome that can be determined from physical and fitness features alone. As such, LightGBM has benefited greatly from GBDT's ability to recognize complex relationships between different variables (i.e., the relationship of low FGC fitness and SDS sleep score). Additionally, linear models (Ridge & Lasso) have shown significant inferiority relative to other models, showcasing that simple additive mappings from physical features to internet-use severity are insufficient.

What these QWK scores mean in practice. A QWK score of .37 represents moderate correspondence between predicted and actual severity classifications. This corresponds to significantly higher agreement than would be observed if randomly assigning a classification (Chance $\approx$ 0.00). However, while the stacking ensemble does accurately identify individuals that fall under either "None", "Moderate / Severe"; it often misclassifies those individuals falling under intermediate severity categories (e.g., predicts

"None" when truth is actually "Mild").

5. Classification

5.1 Pipeline Architecture

The classification pipeline follows the regression pipeline; however, there are two key distinctions:

1. **Class weighting:** a 'class_weight' equal to "balanced" was implemented with lightgbm, XGBoost (through compute_sample_weight), Random Forest, Ridge, SVC, and Logistic Regression. CatBoost utilizes auto_class_weights = 'balanced'. The PyTorch classifier implements class imbalance through internal computation of class weights;
2. **Metric:** using 5-fold stratified kfold cv, **macro f1** score is optimized instead of the QWK metric.

The feature matrix of the classification pipeline utilized 46 columns (45 predictors + is_outlier flag), whereas the feature matrix of the regression pipeline had 45 columns. The is_outlier column is maintained as a binary predictor variable.

IMPUTER_PARAMS (active): {'imputer_choice': 'mice', 'mice_max_iter': 6, 'mice_initial_strategy': 'median', 'mice_et_estimators': 19, 'mice_et_max_depth': 5}

5.2 Hyperparameter Tuning — Optuna Results

Model	Trials	Key search space	Best CV Macro F1
LightGBM	50	learning_rate [1e-3, 0.1], num_leaves [20, 100], max_depth [3, 10], L1/L2 log-uniform	0.3571
XGBoost	50	learning_rate [1e-3, 0.1], max_depth [3, 10], min_child_weight [1, 10], L1/L2 log-uniform	0.3567
CatBoost	50	learning_rate [1e-3, 0.1], depth [4, 10], l2_leaf_reg log-uniform, iterations [50, 300]	0.3496
Random Forest	50	n_estimators [50, 300], max_depth [3, 20], min_samples_split/leaf [1, 20], max_features [0.1, 1.0]	0.3600
RidgeClassifier (+PCA)	100	alpha [0.1, 100], pca_n_components [5, 100]	0.3040
SVC (+PCA)	50	C [0.1, 100], kernel ∈ {linear, poly, rbf, sigmoid}, gamma ∈ {scale, auto}, PCA	0.3282
Logistic Regression (+PCA)	100	C [0.01, 100], penalty = l2, solver ∈ {lbfgs, ...}, PCA	0.3095
PyTorch MLP (+PCA)	100	n_layers [1, 3], units [32, 256], dropout [0, 0.5], lr log-uniform, epochs [50, 500], optimizer ∈ {adamw, adam, sgd, rmsprop}	0.3455

Random Forest achieves the highest tuning CV Macro F1 (0.3600), performing better than LightGBM (0.3571) and XGBoost (0.3567). Linear models (Ridge & Logistic Regression) were somewhat poorer performing models but still useful as baselines.

5.3 Final Evaluation on Test Set.

All classifiers are evaluated on the 1,692-row hold-out test set after full retraining on the training pool (excluding outliers). A simple majority-class dummy classifier (one that always predicts class 0) can be expected to perform at an accuracy rate around 69%, yet a macro f1 score of about 0.17 due to its inability to generate precision/recall/F1 values greater than zero for 75% of the classes. As such, the improvement represented by the models' macro F1 scores from 0.31 to 0.38 represents a notable increase in the ability of these models to provide accurate predictions of a child's sii, although their accuracy rates are lower than the trivial baseline:

Model	Accuracy	Macro F1	Macro AUC
LightGBM	0.59	0.36	0.7082
XGBoost	0.63	0.38	0.7126
CatBoost	0.62	0.37	0.7166
Random Forest	0.58	0.37	0.7222
RidgeClassifier	0.51	0.31	0.6503
Logistic Regression	0.50	0.32	0.6759
SVC	0.52	0.32	0.6949
TorchClassifier (MLP)	0.58	0.35	0.6822

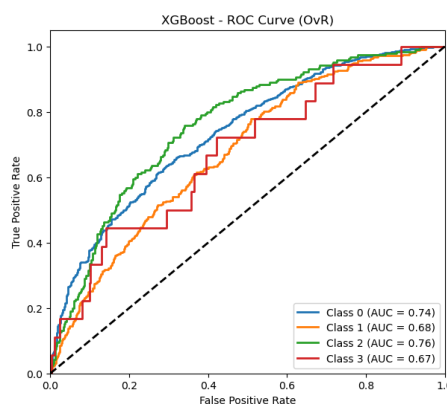
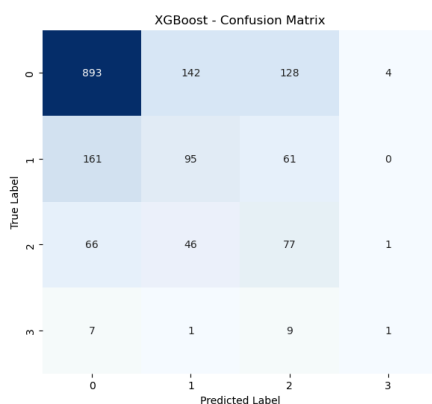
Stacking ensemble (RF + XGBoost + SVC + MLP, meta-learner = Logistic Regression, Nelder-Mead probability multiplier optimisation):

	Precision	Recall	F1-score	Support
Accuracy			0.67	1692
Macro avg	0.35	0.34	0.34	1692
Optimal Class Multipliers	0.5365	1.1254	1.2046	1.0380

Log Loss	0.7724
Quadratic Weighted Kappa	0.3278

It is worth noting that the stacking ensemble optimizes macro F1 scoring (instead of accuracy across the four severity levels), and up-scales the minority-class probabilities (i.e. Multipliers for classes 1 & 2 > 1), while down-scaling class 0 probability (i.e. Multiplier = 0.54). However, class 3 (“severe”) receives zero recall, which makes sense given its extremely low frequency within the test dataset (n = 18): this would mean these 18 children with the most urgent situation would receive no alert at all. Even when utilizing class weighting, there is limited information available for use during training to accurately learn the relationships that exist between the fitness variables and sii severity.

XGBoost (best by Macro F1 and Macro AUC on individual models):



Gradient boosting methods utilizing tree-based structures (XGBoost, CatBoost, Random Forest) outperform both Linear and kernel-based methods consistently. This is consistent with our findings from the regression task, which indicates that the relationship between fitness features and sii exists non lineraly.

Both class 3 (“Severe”) and class 2 (“Moderate”) were difficult to predict for all models. Due to only having 70 examples within the training dataset and highly skewed class weights, classifiers default to treating class 3 as noise. Class 2 was also systematically under-recalled, although some models (CatBoost, RF, TorchClassifier) did achieve recall values ≥ 0.39 and 0.54 for this class.

In addition to poorer threshold calibration than individual base learners, the stacking ensemble experienced a drop in its macro F1 compared to the best performing individual base learner (XGBoost: 0.38 vs. Stack: 0.34) due to multiple structural causes beyond threshold calibration. First, the error patterns of the four base learners (RF, XGBoost, SVC, mlp) are extremely correlated, all four struggle upon the same minority-class instances because they share the same imputed feature space and class-weighted training objective, leaving the Logistic Regression meta-learner with little complementary signal to exploit. Second, the number of positive examples for classes 2 and 3 within the out-of-fold (OOF) predictions used during training of the meta-learner are very low (approximately 150 per fold for class 2, and approximately 14 per fold for class 3), making the second stage learner prone to overfitting upon those layers. Third, the calibration of the multiplier optimization applied during training does not generalize perfectly to the test set, and this is a known limitation of post-hoc threshold tuning without a separate calibration set. In contrast, the stacking ensemble was successful within the regression task due to continuous QWK targets being less sensitive to the per-class calibration issues associated with each of these factors.

A specific comparison across classifiers.

Logistic Regression and Ridge classifier were weaker overall (Macro AUC ~ 0.65 and 0.68), and this reflects their inability to capture interactions between each severity case through projected Linear boundaries via PCA. SVC (Macro AUC 0.69) was able to outperform Ridge although using PCA likely due non-Linear structure through use of a kernel function. The MLP (Macro AUC 0.68) was high but needed careful tuning of PCA parameters;

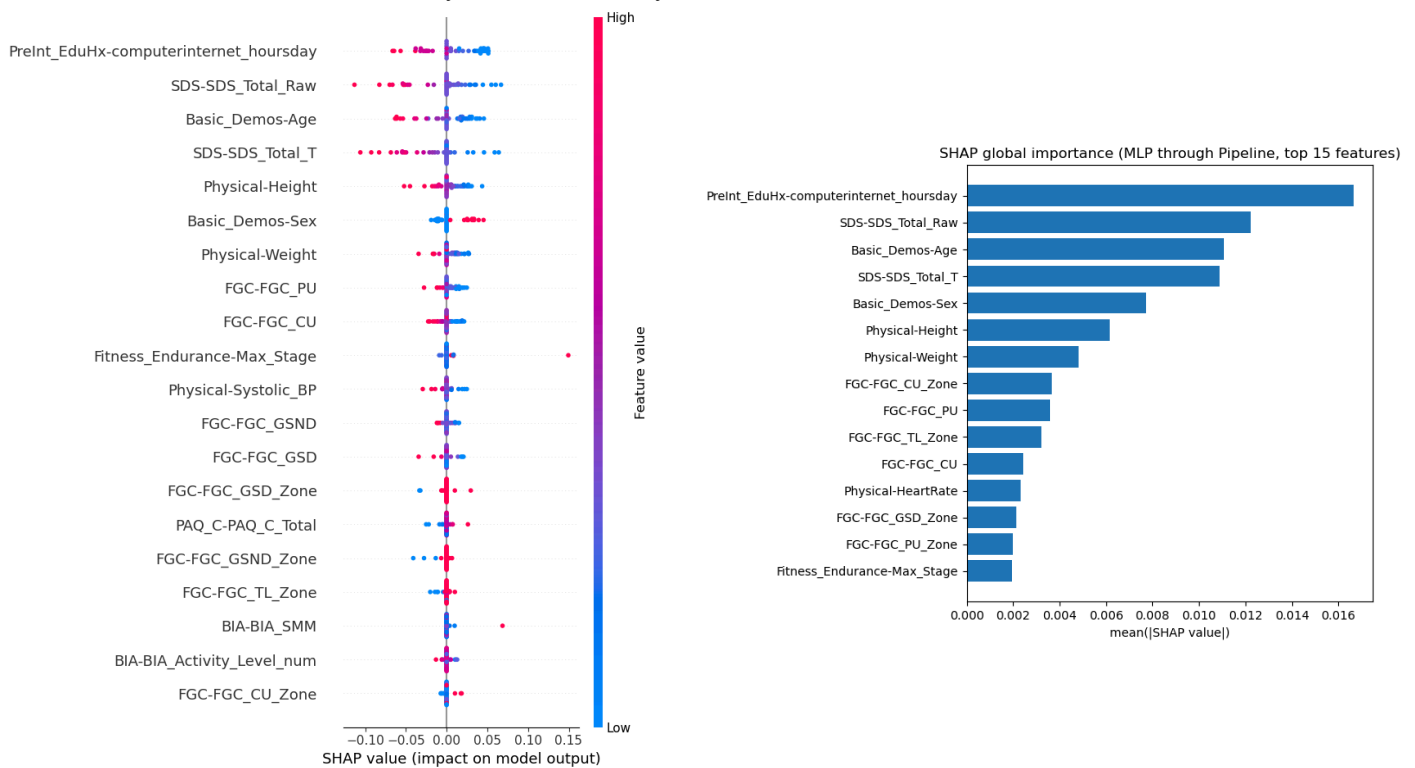
6. Explainability (XAI)

All of the XAI techniques were run on the TorchClassifier (MLP) pipeline on the **pre-PCA** feature space (X_{train_scaled} , X_{test_scaled}) based on four subjects with their respective sii : 0 ($sii = 0$), 5 ($sii = 0$), 10 ($sii = 1$), 15 ($sii = 2$). The MLP was chosen as the basis for the explanation, since there is no built-in native feature-importance mechanism within neural networks like decision trees, and therefore a post explanation is required due to their inherent lack of transparency. However, the MLP is not the highest performing classifier (XGBoost performs better in terms of Macro F1 and AUC). This represents a limitation: when trying to explain a less-than-best model we may assign some degree of importance to certain features that the best model will weigh differently. We point out this consideration to justify the use of MLP based explanations just as an illustration of what can be expected regarding the general feature-importance landscape, rather than definitive assignments of the importance of features to the best-performing pipeline (XGBoost).

6.1 SHAP — Shapley Additive Explanations

Setup. A `shap.KernelExplainer` is created using a set of 50 randomly sampled training examples (`shap.sample(X_train_scaled, 50)`) as background samples. SHAP values are generated for global importance over the first 50 test instances (`nsamples=100`); for local force plots, SHAP values are generated for each of the four individual test cases. The explainer is model agnostic and generates attributions via the `predict_proba` endpoint of the Pipeline in the original 46 dimensional feature space (prior to PCA).

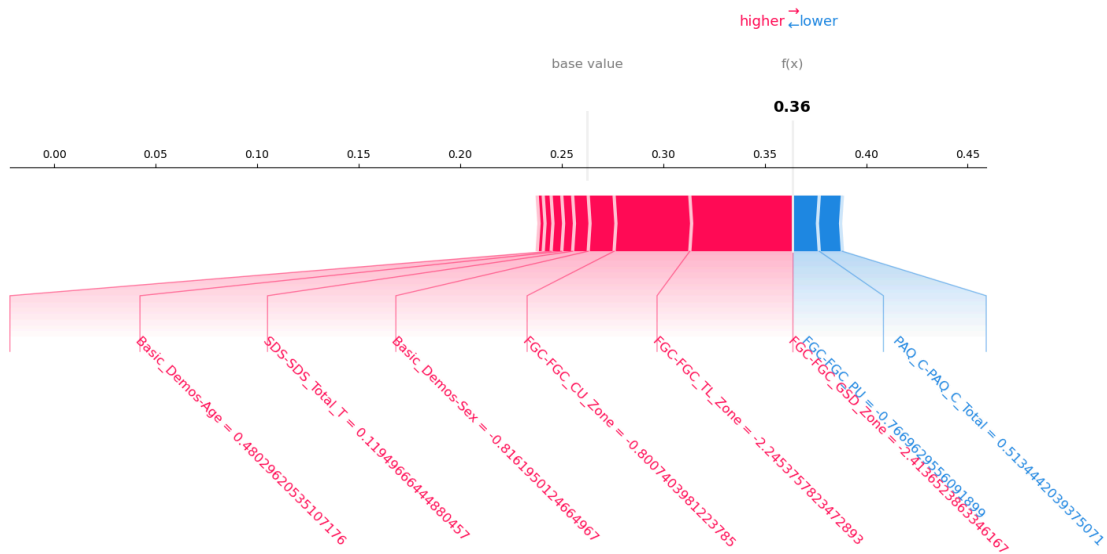
Global importance. The global bar plot (mean $|\text{SHAP value}|$ across all four classes) illustrates that SDS-SDS_Total_Raw, SDS-SDS_Total_T, PreInt_EduHx-computerinternet_hoursday, and several of the FGC fitness metrics contribute to the MLP's decisions. Clearly, sleep disturbance scores are contributing the most, SDS-SDS_Total_Raw and SDS-SDS_Total_T (the z-scores transformed version) are both among the most important predictors. These results are consistent with the previous EDA findings that SDS increases monotonically with severity.



Sleep disturbance as a screening gateway. There is one direct practical implication of the fact that SDS dominates SHAP (and also LIME, TREPAN, and LORE): sleep disturbance screening may be during child development an earlier indicator of impairment, thus children showing normal sleep habits should be classified as being low risk.

Local explanations. The MLP's predictions for each of the four test instances are broken down:

- *Instance 0 ($sii=0$):* The prediction of "None" is supported by low SDS values and a moderate number of hours spent using computers/internet daily, although this is partially offset by physical fitness zone scores that are less than average.
- *Instance 5 ($sii=0$):* Younger participant age and low SDS support the prediction of "None".
- *Instance 10 ($sii=1$):* Elevated SDS values and decreased physical fitness drive the prediction upward towards "Mild".
- *Instance 15 ($sii=2$):* High SDS values and greater than average hours spent using screens daily result in a prediction of "Moderate". (picture below)

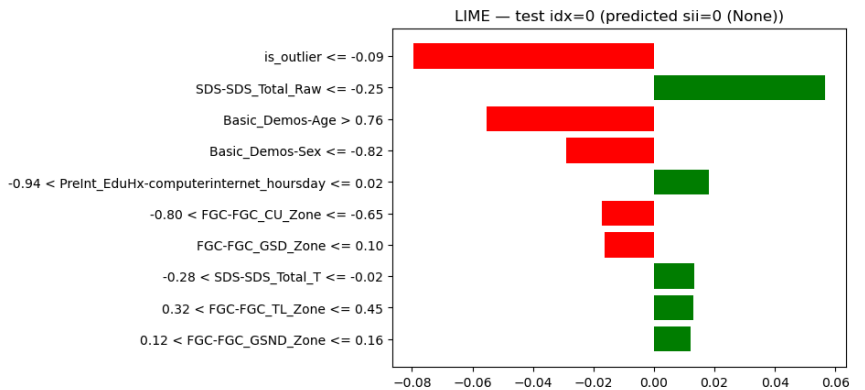


6.2 LIME — Local Interpretable Model-agnostic Explanations

Setup: A LimeTabularExplainer is used to create an explainer object for scaling data with discretize_continuous=True and random_state=42, explaining the MLP's predict_proba function output. An explanation is provided for each test case with num_features=10.

Instance 0 (sii=0):

Feature condition	Weight
is_outlier ≤ -0.09	-0.0797
SDS-SDS_Total_Raw ≤ -0.25	+0.0567
Basic_Demos-Age > 0.76	-0.0554
Basic_Demos-Sex ≤ -0.82	-0.0289
-0.94 < PreInt_EduHx-computerinternet_hoursday ≤ 0.02	+0.0181
-0.80 < FGC-FGC_CU_Zone ≤ -0.65	-0.0171



Low SDS levels (scaled ≤ -0.25) strongly support the "None" classification. Low SDS levels correspond to raw SDS scores less than 39. The age condition Basic_Demos-Age > 0.76 corresponds to approximately 13 years old or more. Finally, the PreInt_EduHx-computerinternet_hoursday ranges from -0.94 to 0.02 in scaled units, corresponding to approximately "Less than 1 hour/day", "Around 1 hour/day" etc. The negative weight assigned to is_outlier ≤ -0.09 is a LIME artifact caused by discretizing this binary feature — this feature is always equal to zero for non-outlier rows, and thus serves as an indication of whether a row lies outside of the normal region.

Instance 10 (sii=1):

FGC-FGC_TL_Zone (trunk-lift fitness zone) show positive contributions toward classifying Instance 10 as having a mild level of internet addiction. Elevated SDS values also support the prediction of "Mild". The trunk-lift fitness zone scores are acting as physical-fitness proxy variables whose decline correlate with mild internet addiction.

Instance 15 (sii=2):

SDS-SDS Total T > 0.04 (Elevated Sleep Disturbance T-Score) is the largest positive predictor (+0.0500). In practice, when we have a T-score in the scaled space above .04 SD it means our raw SDS T-score is somewhere around 58 – 59 (mean = 58.1, SD = 12.6), which is at or just above the population mean, suggesting that even modestly elevated sleep disturbance, not necessarily a clinical-level score, is sufficient to push the model toward a "Moderate" prediction. That is exactly what we see in diagnostic research, where there is a direct relationship between moderate PIU (Problematic Internet Use) and poor sleeping habits. And that directly supports our observation that the SDS (Sleep Disturbances Scale) threshold for being concerned is lower than the therapeutic threshold for a formal sleep disorder diagnosis.

6.3 TREPAN — Global Decision-Tree Surrogate

Setup: TREPAN creates a decision tree surrogate model that mimics the behavior of the MLP oracle. The original dataset is expanded with 500 new synthetic queries created using a Gaussian distribution centered on the original dataset. The oracle labels all synthetic queries. Next, a `DecisionTreeClassifier(max_depth=5, class_weight='balanced')` is trained on both the original and synthetic labeled datasets.

Results:

Surrogate fidelity to MLP on test set: 0.6371 (63.7% agreement)

Surrogate accuracy on true y_test: 0.4397 (43.97% accuracy)

Top-level TREPAN rules (first 3 levels):

[Weight ≤ 1.25] $\rightarrow$ [Age $\leq -0.21 \rightarrow$ (SDS $\leq 0.37 \rightarrow$ (PU ≤ -0.25), SDS $> 0.37 \rightarrow$ (FFMI ≤ 0.13)),

$\rightarrow$ Age $> -0.21 \rightarrow$ (CU_Zone $\leq -0.19 \rightarrow$ (TL_Zone), CU_Zone $> -0.19 \rightarrow$ (SDS_Total))]

[Weight > 1.25] $\rightarrow$ [GSND_Zone $\leq 0.09 \rightarrow$ (Result), GSND_Zone $> 0.09 \rightarrow$ (BMI)]

Interpretation. Physical Weight is the first split made by the surrogate tree based on the fact that the MLP uses weight as a deciding factor for splitting lean participants (i.e. those who weigh no more than 1.25 standard deviations away from the mean, or approximately 134 lbs, based on a mean of 84 lbs and SD of 40 lbs for the entire sample), versus those who are overweight. If a person weighs less, their age is used next as a refining tool to make a prediction, young people (those whose age is 9.5 years or younger, based on a mean of 10.2 and SD of 3.6) are separated by SDS scores regarding their sleep, whereas older individuals are separated based upon their fitness scores (FGC-FGC_CU_Zone).

Developmental note. The age separation occurs at approximately 9.5 years; and this represents the beginning of early pubescence as well as the beginning of increased independence for youths concerning digital media. When considering children who are too young to experience puberty (and therefore do not possess enough independence to access digital media independently), sleep quality seems to be a much stronger indicator of risk; consistent with findings showing that exposure to screens prior to puberty results in the greatest negative impact.

The surrogate model predicts with an accuracy rate of 63.7% (accuracy of agreement with MLP) indicating that a tree with a maximum depth of five captures nearly all, but certainly not all, of the decision-making rules employed by the MLP. The surrogate model has an accuracy rating of 43.97% when considering actual classification labels; which is less than that of a simple majority vote classifier (a classifier that always chooses class zero produces approximately a 69% accuracy rate). However, a classifier employing a simple majority vote achieves an average recall of 0.25; meaning it identifies 100% of cases classified as zero, yet fails to identify any instances of any other classifications. Therefore, although the surrogate model may produce fewer correct classifications, it demonstrates superior overall performance across all classifications.

6.4 LORE — Local Rule-based Explanations

Setup. An 800 sample neighborhood centered around each instance, with $\sigma=0.5$ in scaled feature space, was produced. Each neighborhood was labeled by the MLP oracle. A `DecisionTreeClassifier(max_depth=4, class_weight='balanced')` was trained on each neighborhood. The surrogate tree model's firing rule (the path from root to leaf node that contains the target instance) and one counterfactual (the shortest path to leaf node with alternative classification label) were extracted for each instance:

Firing rules and counterfactuals:

Instance 0 (sii=0, fidelity=0.599):

- **Rule ($\rightarrow$ class 0):** FGC-FGC_TL_Zone ≤ 1.13 AND SDS-SDS_Total_T ≤ -0.17 AND SDS-SDS_Total_Raw ≤ -0.16 AND PreInt_EduHx-computerinternet_hoursday ≤ 0.43
- **Counter-factual ($\rightarrow$ class 2):** same trunk lift zone and SDS conditions, but computerinternet_hoursday > 0.43

The counterfactual shows that if this individual increases their number of daily hours spent accessing internet above .43 SD units above the mean (approximately equivalent to responding with an ordinal value of 1.4 on a scale where responses range from approximately "about 1 hour/day", to "about 2 hours/day"), they will move into the "moderate" category according to the MLP's local reasoning. In reality, 0.43 SD above the mean of preint_eduhx-computerinternet_hoursday (which has a mean of approximately 1.0 and a standard deviation of approximately 1.0) corresponds to an approximate value of 1.4 on an ordinal scale (that ranges from "about 1 hour/day" to "about 2 hours/day"). However, the local fidelity for this instance is only 0.599, indicating that the local surrogate tree agrees with the MLP oracle on only about 60% of the perturbed neighbourhood. This rule should therefore be interpreted with caution and not taken as a reliable indication of the exact threshold.

Instance 5 (sii=0, fidelity=0.820):

- **Rule:** SDS-SDS_Total_Raw ≤ -0.05 AND PreInt_EduHx-computerinternet_hoursday ≤ 0.49
- **Counter-factual ($\rightarrow$ class 2):** low SDS but computerinternet_hoursday > 0.49 AND SDS-SDS_Total_T > 0.09

Both instances 0 and 5 share the structure: normal (or low) sleep disturbance plus limited daily screen time classify as "None". The tipping point in both counterfactuals is an increase in `computerinternet_hoursday`, confirming that the model treats screen-time self-report as a reliable separator. With a local fidelity of 0.820, this is the most trustworthy of the four LORE explanations. Concretely, this means that, holding other factors constant, a child whose daily screen time rises above approximately 1.5 hours per day (0.49 SD above the mean of the ordinal screen-time scale, corresponding roughly to the boundary between "around 1 hour" and "around 2 hours") would be predicted to shift from "None" to "Moderate".

Instance 10 (sii=1, fidelity=0.635):

- **Rule ($\rightarrow$ class 2):** `PreInt_EduHx-computerinternet_hoursday > -1.02 AND FGC-FGC_CU_Zone  $\leq$  1.55 AND SDS-SDS_Total_T > 1.46 AND FGC-FGC_TL_Zone  $\leq$  0.76`
- **Counter-factual ($\rightarrow$ class 0):** `computerinternet_hoursday  $\leq$  -1.02 AND SDS-SDS_Total_Raw  $\leq$  2.24`

Note: the local tree predicts class 2 for this neighbourhood region, whereas the global MLP predicts class 1. The fidelity (0.635) reflects this discrepancy between local surrogate and oracle. The SDS condition `SDS-SDS_Total_T > 1.46` translates to a raw T-score above approximately 76 (mean $\approx$ 58.1, SD $\approx$ 12.6; $1.46 \times 12.6 + 58.1 \approx 76.5$), which is well above one standard deviation from the mean and would be considered diagnostically elevated. Because the local fidelity is below 0.70, the counterfactual rule for this instance should be interpreted as approximate rather than as a reliable intervention threshold.

Instance 15 (sii=2, fidelity=0.762):

- **Rule ($\rightarrow$ class 2):** `BIA-BIA_BMC  $\leq$  0.80 AND SDS-SDS_Total_Raw > -0.54 AND SDS-SDS_Total_T > 0.10 AND computerinternet_hoursday > -0.67`
- **Counter-factual ($\rightarrow$ class 0):** `BIA-BIA_BMC  $\leq$  0.80 AND SDS-SDS_Total_Raw  $\leq$  -0.54 AND SDS-SDS_Total_T  $\leq$  0.50 AND FGC-FGC_CU  $\leq$  0.92`

`BIA-BIA_BMC` (Bone Mineral Content, a bioimpedance-derived body composition metric) appears in the counterfactual: normalising sleep disturbance below -0.54 SD would shift the prediction from "Moderate" to "None". In real-world terms, the SDS condition `SDS-SDS_Total_Raw  $\leq$  -0.54` corresponds to a raw SDS score below approximately 36 (mean $\approx$ 41.4, SD $\approx$ 9.5; $-0.54 \times 9.5 + 41.4 \approx 36.3$), which is well within the normal range. The BIA feature's presence is likely a proxy for overall body composition and physical development stage rather than a direct causal factor. With a fidelity of 0.762 (above the 0.70 threshold), this counterfactual is reasonably reliable. Concretely, it suggests that, holding body composition constant, a child whose sleep disturbance score falls below 36 (approximately 0.5 SD below the population mean) would be predicted to shift from "Moderate" to "None". For a clinician, this reinforces the centrality of sleep quality as an intervention target: even moderate improvements in sleep hygiene may be sufficient to move a child out of the model's "Moderate" risk category.

7. Tabular Analysis Discussion and Conclusions

7.1 What the Models Learned

The common top predictors in the regression and classification pipelines include:

1. **Sleep Disturbance (SDS-SDS_Total_Raw / SDS-SDS_Total_T)**, the largest contributor in the feature space. Each of the XAI methods independently point to SDS as the greatest contributor and there is a significant body of medical evidence suggesting that children using screens excessively experience disruptions to their sleep.
2. **Daily internet hours (PreInt_EduHx-computerinternet_hoursday)**, an independent, although very coarse, measure of time per day spent on the internet. LORE counterfactuals indicate this is the key variable differentiating 'none' and 'moderate' in borderline instances. In absolute terms, the cut-off for risk seems to exist somewhere between around 1 hour per day and 2 hours per day on the ordinal scale.
3. **Fitness metrics (FGC zone scores)**, Curl-up zone (FGC-FGC_CU_Zone, binary: 0 = Needs improvement, 1 = Healthy fitness zone) and Trunk Lift zone (FGC-FGC_TL_Zone, identical coding) occur frequently in the explanations produced by SHAP, LIME, TREPAN, and LORE. Children exhibiting poorer cardiovascular and muscular fitness exhibit greater severity.
4. **Body composition (Physical-BMI, Physical-Weight, BIA-BIA_BMC)**, secondary contributors which probably relate to sedentary behavior instead of causing the behavior directly.

All of the reported models omit the PCIAT features and thus identify physical world correlates of digital behavior, a much more useful and more generalizable result than reconstructing a version of the PCIAT test itself.

Sleep disturbance as a universal screening gateway. The fact that all four XAI techniques identified SDS as the major contributing factor to the predictive model has implications for practice. Therefore, a two stage screening process is feasible and actionable: (1) Screen all children in the appropriate age range for sleep disturbances, (2) Only administer the full fitness battery and further evaluation to those scoring at or above a predetermined threshold (e.g. raw SDS > 45 which is roughly equivalent to the average value of Class 2). This would greatly decrease screening costs while still identifying the most informative predictor as the best pareto efficient solution.

7.2 Limitations

- **Class 3 ("Severe")** remains difficult to predict. No single model is able to provide meaningful recall for this class given the 70 available training samples. Even employing ensembles and optimizing thresholds does little (if not worse) to improve prediction accuracy for Class 3. As noted throughout this report, the cost of failing to detect severe cases is not symmetric with other errors: in a real screening context, these are the children most urgently in need of “medical” intervention.
- **Leakage caveat.** All results exclude the PCIAT features. Thus any comparisons made with leader board scores from Kaggle (which do allow inclusion of PCIAT) are invalid.
- **Imputer selection bias.** The imputer was selected based upon performance against a holdout mask of a portion of the overall training sample, not within a fully nested cross-validation scheme. Therefore the reported cross-validation QWK/F1 should be viewed as somewhat overly optimistic.
- **SHAP KernelExplainer approximation**, at `n_samples = 100` SHAP values approximate. Stabilization of feature importance rankings among lower-importance features may not occur.
- **XAI model choice explanations** were generated for MLP but not XGBoost (best performing classifier). Although this choice is warranted because we wish to demonstrate explanatory power for a full black box model; consequently the feature attribution may not precisely mirror the decision making logic employed by the actual deployed model.

7.3 Summary of Key Results

Track	Best model	Best metric	Ensemble
Regression	LightGBM	QWK 0.3613	Stack QWK 0.3712
Classification	XGBoost	Macro F1 0.38, AUC 0.7126	Stack Macro F1 0.34 (lower)

Regression stacking outperformed any single model. Classification stacking under performed best single model likely due to a variety of reasons including correlation among base learners, over fitting by meta learner on sparsely populated minority class folds, and limitations in calibrating thresholds on test set.

Methodological contributions. The research demonstrated a robust outlier detection consensus (10 detectors, $\geq 9/10$ agreement, 99th-percentile threshold); a controlled imputation tuning procedure; and a completely separate classification-regression pipeline utilizing the same splits of raw data. The XAI section utilized all four required interpretative techniques (SHAP, LIME, TREPAN, LORE) on the same model and instances providing opportunity to compare and validate each method’s assignments of feature importance.

8. Time Series: Introduction

The CMI time series dataset includes records coming from a wrist-worn actigraph; what was recorded are measures to assess physical activity, posture, and environmental factors experienced by subjects over the observation period. Our goal through the analysis of this dataset is to discover if any relation exists between specific movements/lack of movement and unhealthy internet usage, represented by 'sii_binary'. More specifically, a `sii_binary = 0` represents non-problematic internet use, while `sii_binary = 1` represents problematic behaviors.

The dataset considers 4,437 subjects, and for each subject, we record data deriving from their wrist sensors over 200 time stamps. Each time stamp is 30 minutes apart (making the entire recording period roughly 4 days).

This final section about time series analysis will encompass appropriate preprocessing, clustering, matrix profile to discover motifs and discords, and finally classification.

8.1 Time Series Data Understanding & Preparation

Dataset Structure and Test-Train Split.

There are a total of 13 columns in the dataset, which can be summarized in the table below.

Channels	Description
'X', 'Y', 'Z'	Raw tri-axial accelerometer readings (wrist orientation)
'enmo'	Euclidean Norm Minus One, the calibrated acceleration proxy for overall physical activity intensity (in g). An 'enmo' of 0 indicates that the subject is completely still
'anglez'	Wrist angle relative to horizontal, used as proxy for posture and sleep/wake state (in degrees)
'non-wear_flag'	Binary indicator of whether the device was being worn

Channels	Description
'light'	Ambient light sensor, distinguishes indoor/outdoor and time-of-day context (in lux)
'battery_voltage'	Battery status of the device (in mV)
'weekday',' quarter', 'relative_date_PCIAT'	Time features
'id'	Unique subject id
'sii_binary'	Binary value for the Severity Impairment Index

The features 'weekday',' quarter', and 'relative_date_PCIAT' have been removed before performing the train-test split, as they were not deemed crucial for the classification task. It is worth mentioning that during data preprocessing, 'non-wear_flag' has not undergone outlier detection or standardization, as binary features need neither of these checks to be performed.

A vertical split has been performed: on one side, a 80-20 train-test split has been performed only on 'sii_binary', leading to the creation of 'y_train' and 'y_test'; the remaining 8 features have been subject too to a 80-20 train-test split and divided into 'x_train' and 'x_test'.

The dataset is heavily imbalanced: of the total 4,437 subjects whose data was recorded, a total of 2,957 people (roughly 66.6%) are categorized as non-problematic, and the remaining 1,480 (approximately 33.4 %) are categorized as problematic. The test and train splits have been made in a way which ensures this class imbalance has been kept also after the splits.

Data preprocessing: Missing Values, Outliers, and Standardization.

Missing values.

A check for missing values has been performed, and the results confirm that there is none in the dataset.

Outlier detection.

The next step was to identify any outliers, and for this task we used Hampel Filter with a window size of 7 and a sigma equal to 4.5; to be specific, we have looked for outliers within each single feature of each respective subject. Two additional columns have been left out of the analysis for this task: 'non-wear_flag' and 'id', as both do not have outliers. We have not looked for outliers in 'y_train' and 'y_test'. The plot below visually summarizes the functioning of the Hampel filter, by showing the detected outliers for the column 'enmo' of Subject 0: it is possible to see which outliers were found, removed, and replaced through linear interpolation.



Standardization.

Standardization allows us to rescale the data, so that it shares a common scale, and comparison becomes easier. What has been done for this section is to use Z-Score Standardization on both 'x_train' and 'x_test'; the choice of this specific method is due to some of the classification models we will be using in later sections (KNN with DTW, Shapelet) working better on data which has undergone such normalization. What this does, is to force every feature to have mean (μ) equal to zero, and standard deviation (σ) equal to 1. Once again, 'y_train' and 'y_test' have been left out, as binary features do not need normalization.

8.2 Time Series Approximation, Clustering, and Matrix Profile.

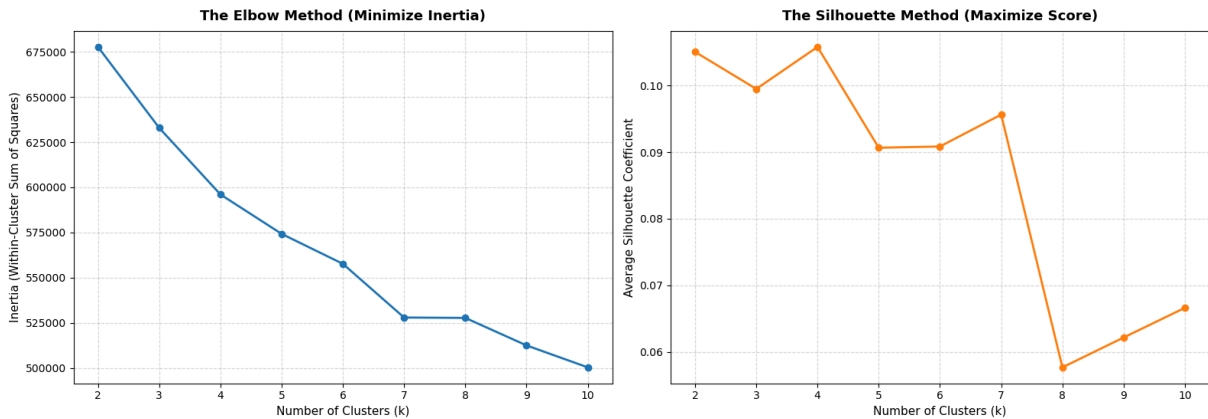
Time Series Approximation.

Due to the large size of the dataset we are working with, Piecewise Aggregate Approximation (PAA) has been performed, with the intention of simplifying the time series while still retaining key structural features and insights. What has been done was to reduce the 200 timestamps to only 40 PAA segments. The clustering algorithm below have been applied exactly on this approximated version of our data.

Clustering.

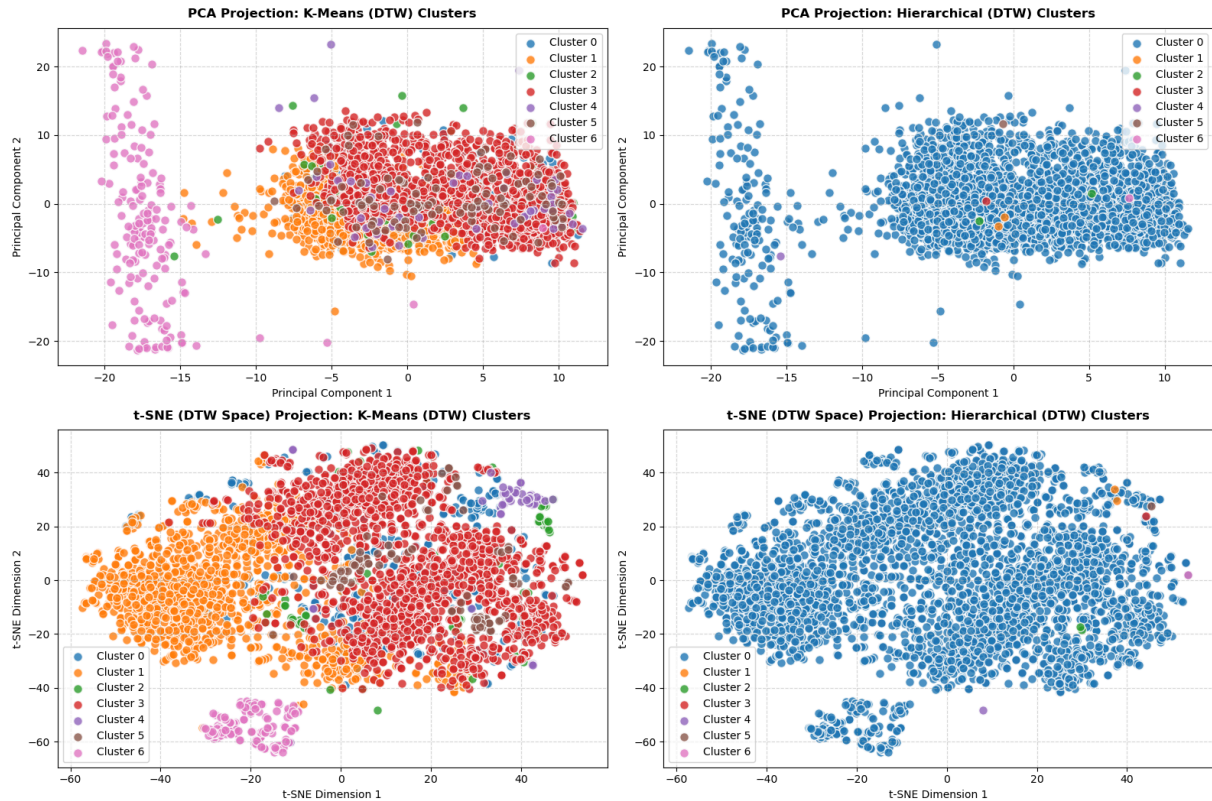
For the clustering tasks two algorithms have been used: K-Means and **Hierarchical Clustering**, both using Dynamic Time Warping (DTW) as distance measure. Computing the DTW distance measure is extremely time consuming. For this reason, we have imposed the Sakoe-Chiba band constraint with radius 4, with the intention of reducing computational time. In order to find the optimal number of clusters for the K-Means algorithm, the Silhouette Method and the Elbow Method have been used, and both revealed 7 to be an appropriate cluster number. This same number of clusters has then also been used for Hierarchical Clustering.

Finding the Optimal Number of Clusters (k)



Both clustering algorithms have been visualized onto a 2D through two dimensionality reduction methods, Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE).

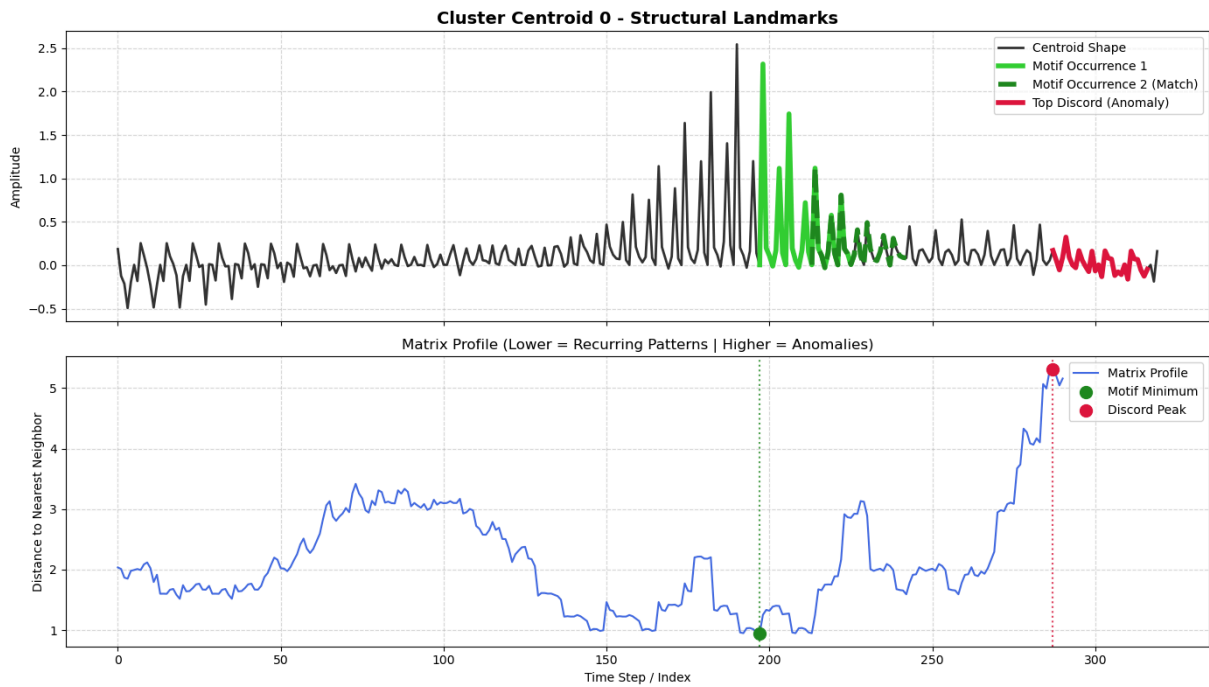
DTW-Based Time Series Clustering Evaluation & Visualization



K-Means seems to be better at distinguishing among clusters with respect to Hierarchical Clustering. For K-Means we can see that with both PCA and t-SNE visualization overall the different clusters are separated, with the exception of Cluster 0. Hierarchical Clustering on the other hand did not perform very well, and the combination of DTW and Average Linkage method seem to have caused a chaining effect, causing nearly all points to be considered of Cluster 0, with few sparse points being part of other clusters.

Matrix Profile: Motifs and Discords Discovery.

We have then moved on to Matrix Profile, which has been performed on the cluster centroids identified by the K-Means algorithm. The window size has been set to 30, and the exclusion zone to find discord is equal to 15. From the plot below, which only shows the Matrix Profile for cluster 0, we can clearly notice the motifs, which look like valleys, and the discords, which are represented by peaks. The top motif found is the one highlighted by the green dot in the second plot, which is exactly in correspondence of the subsequence which starts at index 197. On the other hand, the top discord can be found at Subsequence starts at index 287.



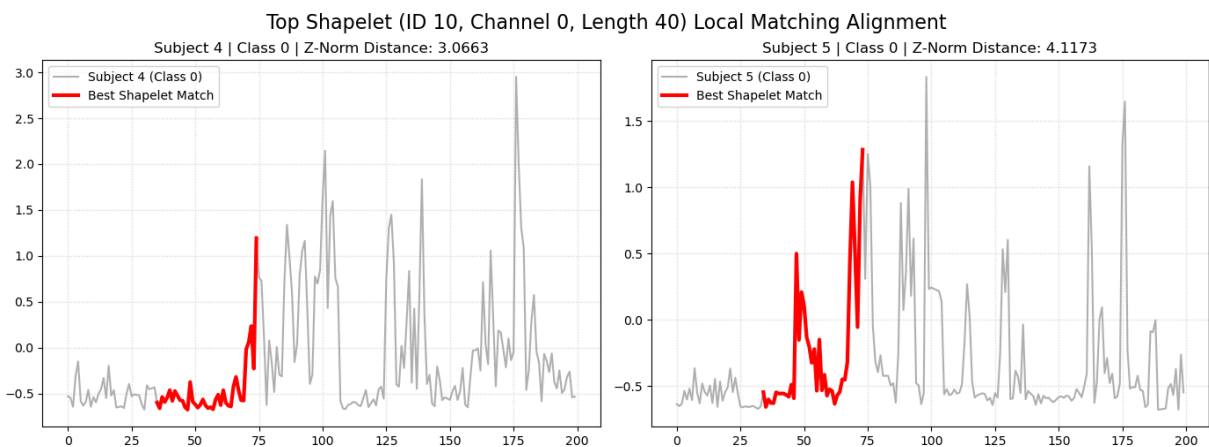
9. Shapelets.

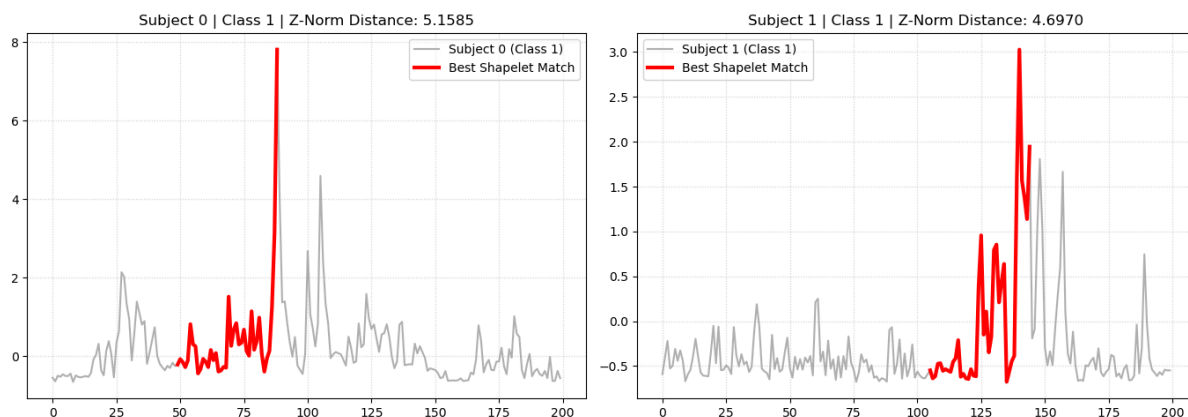
The core difference between Matrix Profile and Shapelet is that through the Matrix Profile we are able to visualize repeating patterns in the time series, but through Shapelet we specifically try to extract patterns, or shapelets, which are highly representative of one class, and then use it to discriminate during the classification task. To speed up our calculations, we made use of both Distance Early Abandon and Admissible Entropy Pruning.

We have generated 15 shapelet candidates, testing through different window lengths: then, for every candidate we saw how well it matched the time series, and finally we used Information Gain to determine how well each shapelet candidate divided our classes; through this method, we were able to prune 7 candidates early on. Shapelet candidate number 10 was the one which reached the highest Information Gain (0.003205), and that had a split threshold of 4.4712.

Additionally, the following factual rule has been extracted: “If Time Series Q contains a subsequence within a distance threshold of 4.4712 matching Shapelet 10, then classify as Problematic (Class 1); else Non-Problematic (Class 0)”.

The plot below shows a visual representation on how well Shapelet 10 is able to distinguish between different subjects, which belong to different classes (Class 0 or Class 1).





10. Time Series Classification

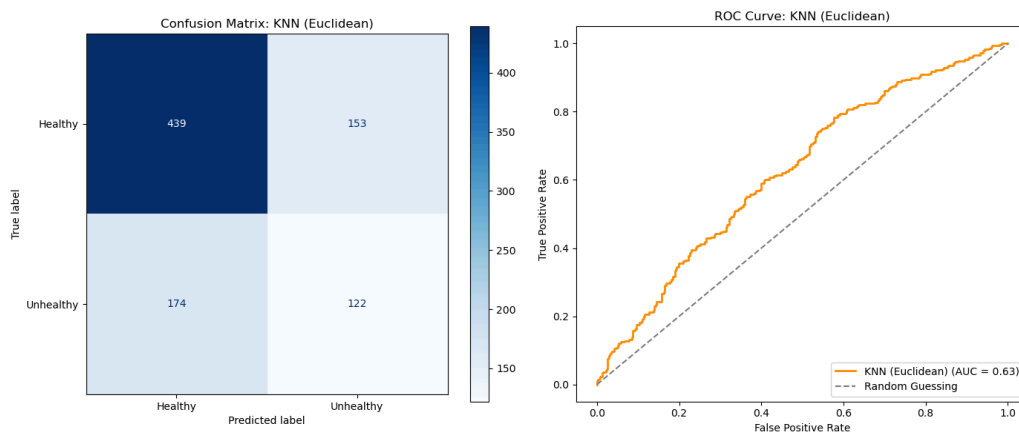
Task Definition and Baseline

In this classification section we will be testing a multitude of models to see which is more able to correctly discriminate between the different 'sij_binary' classes. Since we are dealing with a highly unbalanced dataset, we should keep in mind that any classifier that fails to achieve better than a **naive baseline of 66.0%** has no ability to discriminate between the classes. Accuracy, macro F1 score, and AUC-ROC were used as metrics in which to evaluate all classifiers; each was evaluated by means of 4-Fold Cross Validation for optimizing thresholds.

10.1 Model 1: K-Nearest Neighbor with Euclidean Distance

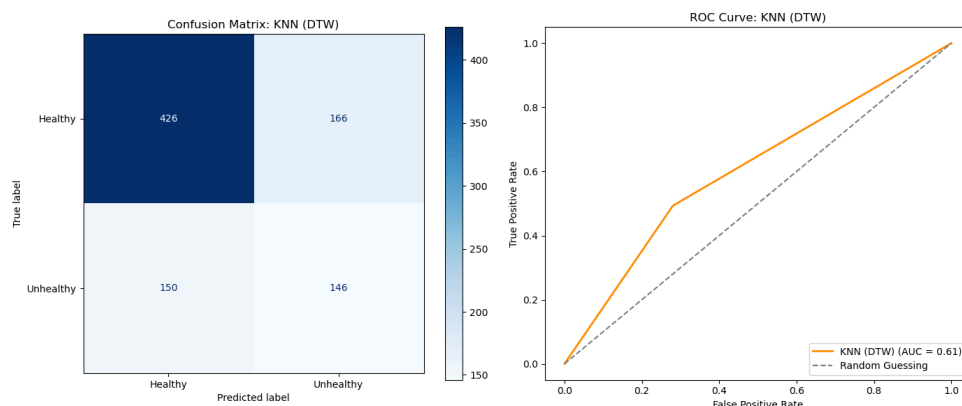
We have started by using the KNN algorithm with Euclidean distance and utilizing Optuna (10 trials, 4-Fold CV, macro F1 objective) to tune hyperparameters for optimal performance; the classifier was optimized using distance weighted voting. In evaluating the classifier on the test dataset, it performed with an accuracy = 0.6318, macro F1 = 0.5780, a weighted F1 = 0.6282, and AUC-ROC = 0.6256.

When looking at the confusion matrix, the typical challenges of dealing with an unbalanced dataset are evident. The classifier achieved a recall rate of approximately 74% for Class 0, and roughly 41% for Class 1. Therefore, nearly two thirds of high risk children were missed due to the imbalance in the data. The ROC Curve suggests that this model has limited potential when compared to random guessing, as the model performs slightly better than random guessing.



10.2 Model 2: K-Nearest Neighbor with DTW

If compared to Euclidean Distance, DTW is deemed superior when working with time series: it works by matching two series by stretching or shrinking one series based upon the other, while minimizing total cumulative distance. A Sakoe-Chiba band of 10% (20 steps) was applied to restrict the optimization path and to keep the process manageable for thousands of multivariate samples.

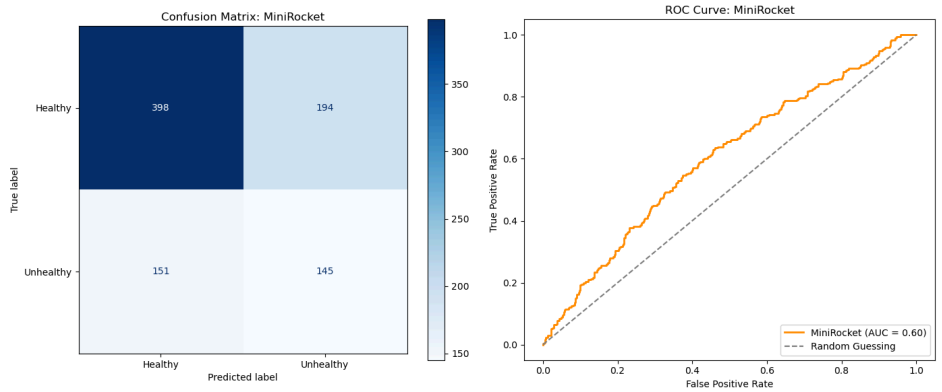


Its inherent theoretical advantage over Euclidean distance can be seen by the model's reported performance: accuracy = 0.6441, macro F1 = 0.6049, AUC-ROC = 0.6064, Weighted F1 = 0.6464.

The KNN-DTW confusion matrix showed better recall for Class 1 compared to KNN-Euclidean (49% DTW vs. 41% Euc), Class 0 precision was two percentage points higher than KNN-Euclidean, and therefore overall accuracy was slightly better. The ROC curve looks better than the previous one.

10.3 Model 4: MiniRocket

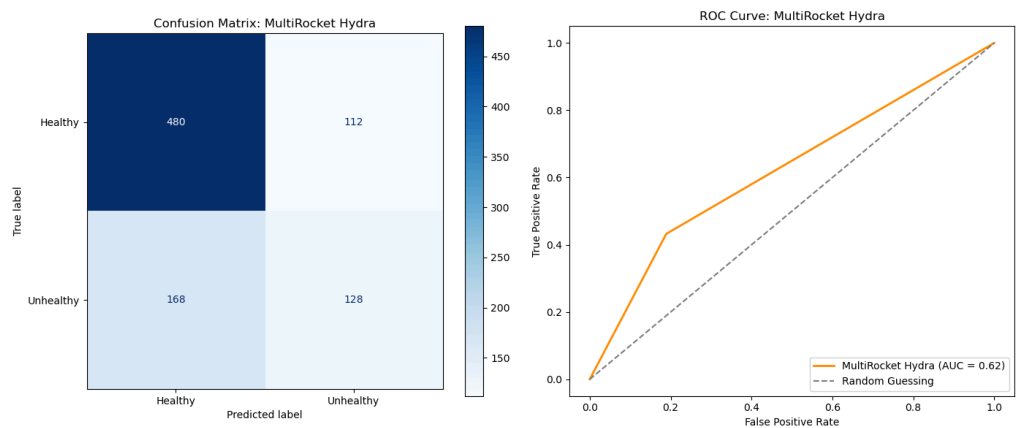
MiniRocket generated a very large number (10,000) of fixed random convolutional kernels and computed the proportion of positive values (PPV) as features which were fed into a logistic regression classifier. The method was developed primarily for its potential in providing fast and competitive results on time-series benchmarks. An improvement in model convergence was made by raising `max_iter` to 5000 and adding the normalization of the features. MiniRocket achieved an accuracy of 0.6115, a macro-F1 score of 0.5772, a weighted F1 of 0.6173, and an AUC-ROC score of 0.6023.



Specifically, the model achieved a 67% recall rate for Class 0, and a 49% recall rate for Class 1. The corresponding ROC curve shows once again, does not show any improvement from the previous two methods.

10.4 Model 4: MultiRocket Hydra.

MultiRocket Hydra combines the multi-feature extraction capabilities of MultiRocket with the approach of the Hydra framework to capture complex, local time-series patterns. To optimize classification performance on this dataset, a decision threshold tuning step was performed during cross-validation, establishing an optimal threshold of 0.100.



MultiRocket Hydra achieved an accuracy of 0.6847, a macro-F1 score of 0.6259, a weighted F1 of 0.6753, and an AUC-ROC score of 0.6216. Specifically, the model achieved an 81% recall rate for Class 0 (Healthy), and a 43% recall rate for Class 1 (Unhealthy). The corresponding ROC curve demonstrates a modest overall discriminative power, though it represents a clear performance step up from the baseline MiniRocket method.

10.5 Model 4: Multivariate PULSAR.

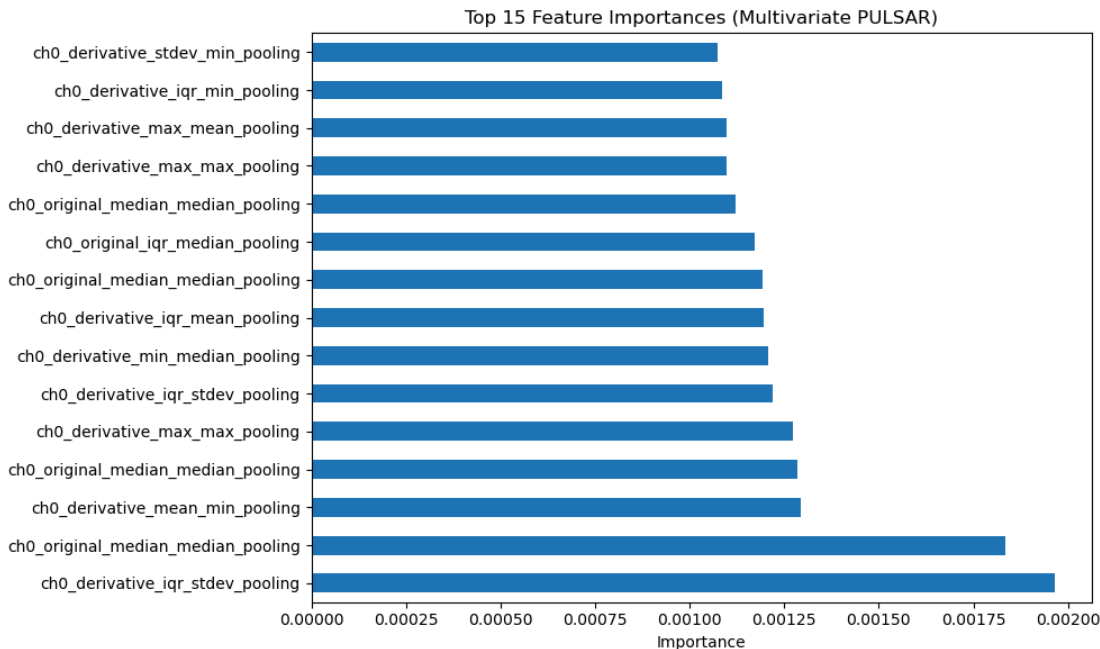
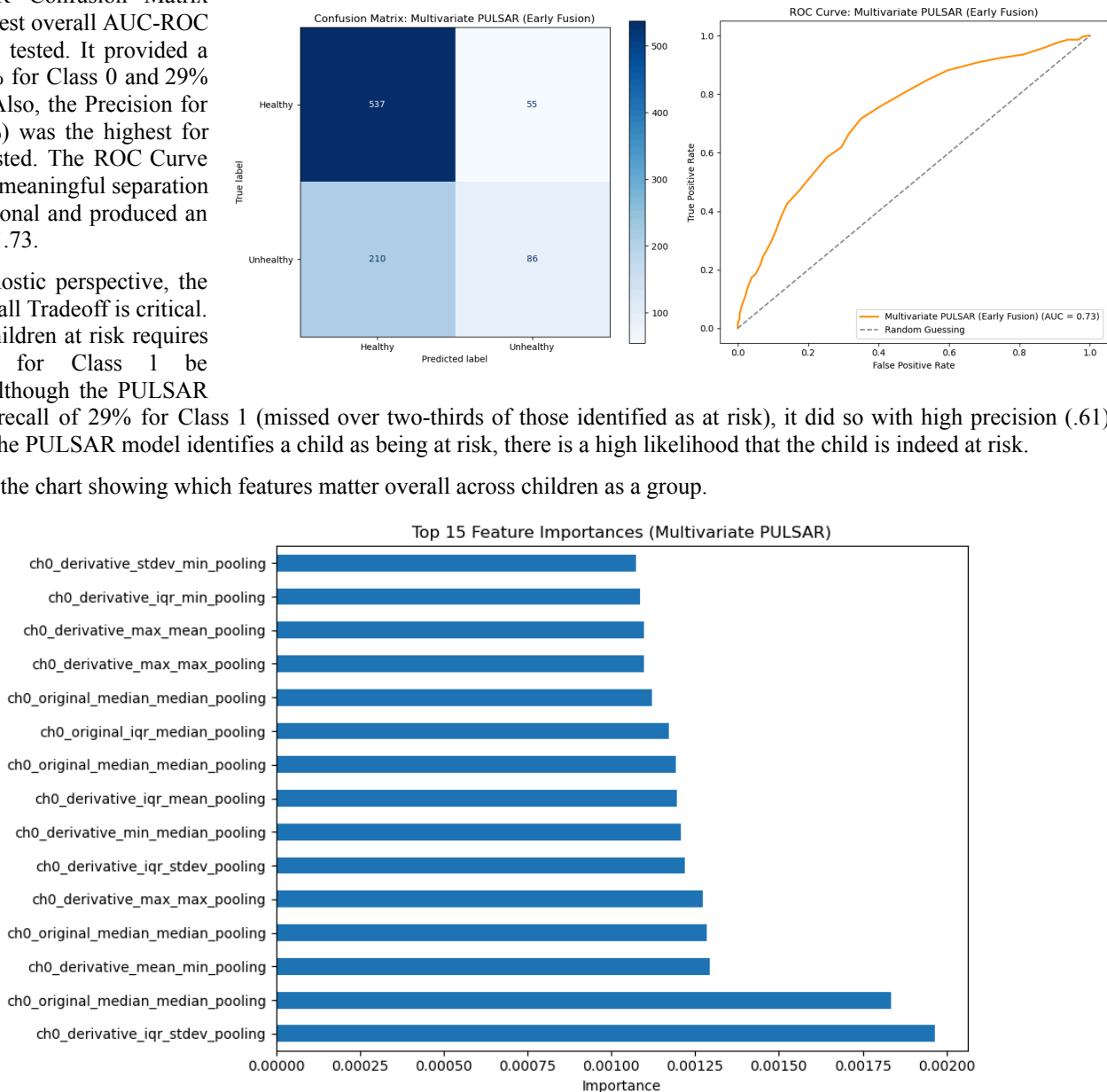
The Multivariate PULSAR model did not simply analyze raw data within a set time frame to determine whether a child had been identified as being at risk. It would have provided no explanation as to why the child had received such designation. I therefore converted each signal into a meaningful feature set (trends; spike patterns; and rhythm patterns), and then allowed an Extra Trees model to learn from those sets. I also employed LORE, allowing me to view the specific explanations behind that particular prediction for every individual child, not merely a single number. The goal was to test whether this methodology could provide true explanatory insights rather than pure speculation.

Pulsar, clearly exceeded our naive baseline. It achieved an accuracy score of .702 as well as a 3.5% increase above majority-class-guessing (naive baseline .667). Additionally, the model showed good discriminative power through a Macro-F1 Score of .598 and an Area under the Receiver Operating Characteristic Curve of .730. This demonstrates how the PULSAR model is able to effectively combine multi-scale representations (i.e., original signal, frequency spectrum, first and second order derivatives, AR Coefficients) using an Extra-Trees Classifier with balanced class weights.

The PULSAR Confusion Matrix provides the best overall AUC-ROC of any model tested. It provided a Recall of 91% for Class 0 and 29% for Class 1. Also, the Precision for Class 1 (61%) was the highest for any model tested. The ROC Curve demonstrated meaningful separation from the diagonal and produced an AUC value of .73.

From a diagnostic perspective, the Precision-Recall Tradeoff is critical. Identifying children at risk requires that recalls for Class 1 be prioritized. Although the PULSAR model had a recall of 29% for Class 1 (missed over two-thirds of those identified as at risk), it did so with high precision (.61). Therefore, if the PULSAR model identifies a child as being at risk, there is a high likelihood that the child is indeed at risk.

Below we see the chart showing which features matter overall across children as a group.



LORE explanations below focus on which features matter for one specific child at a time in this case LORE builds rules based only on that child's local neighborhood. So the specific features it picks (Channel 1 features appear below) don't have to match the top 15 overall list; they just have to separate that specific child from being classified as 'at risk'.

LORE Local rule-based explanations

- test idx=0 | ExtraTrees -> Healthy | local fidelity=0.994
Rule (-> class 0): IF ch0_orignal_iqr_stdev_pooling > -0.26 AND ch1_orignal_max_min_pooling > 0.01

Counter-factual (-> class 1): IF ch0_original_iqr_stdev_pooling <= -0.26 AND ch0_derivative_iqr_min_pooling <= -0.46 AND ch0_original_max_median_pooling > -0.39 AND ch1_original_slope_stdev_pooling <= 0.10

- test idx=5 | ExtraTrees -> Healthy | local fidelity=1.000

Rule (-> class 0): IF ch1_periodogram_stdev_max_pooling <= 0.82 AND ch1_original_max_mean_pooling > 0.49

Counter-factual (-> class 1): IF ch1_periodogram_stdev_max_pooling > 0.82 AND ch0_original_iqr_mean_pooling > -0.69

- test idx=10 | ExtraTrees -> Healthy | Neighbourhood generated only one class, cannot fit LORE tree.

For instance, as shown in Test Case 0, the Model called this Child “Healthy” due to its Activity Pattern remaining consistently above the Stability and Movement Thresholds (i.e., ch0_original_iqr_stdev_pooling > -0.26 and ch1_original_max_min_pooling > 0.01). There was an extremely high degree of similarity between this Child's Behavior and the Rule used to predict the classification (Local Fidelity = .994; therefore, the Rule matched the Model's actual prediction for 99.4% of similarly-simulated Children). The Counter-Factual illustrates what would have changed this classification to “At Risk,” namely a reduction in Variability in the Child’s Activity Patterns and Changes in the Rate of Change (Derivative) Signals.

The Concept illustrated in Test Case 5 follows a similar path, however the characteristics are generated by a set of frequency/periodogram-based features and has an equally confident Local Rule (Fidelity = 1.000).

Test Case 10 provides another excellent illustration of how LORE can fail to produce a classification rule when it cannot find sufficient contrast between neighboring children to provide a comparison to classify the focal child. In other words – this child’s data point exists within such a narrow band of ‘Stable’ data points that do not border on either side (and thus does not exist adjacent to a sufficiently-different “At Risk” pattern) – that no meaningful rule could be derived by LORE.

10.6 Hyperparameter Tuning

Optuna was utilized to perform a hyperparameter search on the KNN-Euclidean model, exploring a search space of $k \in [1, 20]$ across 10 discrete trials. Each trial evaluated a single k-value configuration using 4-fold cross-validation optimized for a macro F1 objective. This optimization framework selected $k=6$ as the best overall configuration, achieving a cross-validation macro F1 of 0.533. We did not do hyperparameter tuning on the following very computationally intensive algorithms (KNN-DTW, RDST, MiniRocket, MultiRocket-Hydra) because repeated DTW calculations are too time consuming or shapelet extraction is too time consuming. As for the PULSAR model, we instead performed its internal calibration (four fold cross validation with a threshold optimization of 0.33).

10.7 Comparative Summary

Model	Accuracy	Macro F1	AUC-ROC	Class 1 Recall
Naive Baseline	0.670	—	—	0.00
KNN-Euclidean	0.6318	0.5780	0.6256	0.41
KNN-DTW	0.6441	0.6049	0.6064	0.49
MiniRocket	0.6115	0.5772	0.6023	0.49
MR-Hydra	0.6847	0.6259	0.6216	0.43
PULSAR (Early Fusion)	0.7016	0.5978	0.7295	0.29

11. Sequential Pattern Mining

11.1 Discretization: PAA + Quantile Binning

The discrete version of the continuous ENMO signal will have to be created in order to sequentially mine the pattern. To this end, a two-stage process was utilized:

First stage: Piecewise aggregate approximation (PAA). PAA took every 200 step series and collapsed it into 20 different series of 10 steps each (5 hour segments; 10 times compression) sequence mining needed coarser 5-hour blocks to capture distinct morning/afternoon/night routines instead of the 3.33 hours used previously in time series clustering.

Second Stage: Quantile Binning. Every time an individual series had been compressed into 20 different numerical segments, which are later turned into symbols from the original series, they were then binned into one of three categories using the global terciles as a threshold. The thresholds are: Low(L) ≤ -0.3673 , Medium(M) $= -0.3673 - 0.0215$, High(H) ≥ 0.0215 .

Three symbols were used to provide enough expression for capturing useful patterns and prevent excessive fragmentation of support for those patterns. Too few symbols may cause loss of important distinctions among classes, while too many symbols would create excessive fragmentation of support for each class among all possible patterns. Physiologically, low represents resting or sleeping activities, medium represent very mild physical activity (e.g. walking, performing household chores), high represents moderate to vigorous physical activity. This provided an equal division of all the possible states (symbol distribution) globally at 33.3%. Thus, the intent of creating a uniformly divided space through the use of tercile binning was accomplished. Crucially, class-conditional symbol distributions revealed a first discriminative signal: problematic children produced 38.4% L symbols versus 30.8% for non-problematic children, while H symbols showed the inverse pattern (26.7% vs. 36.6%)

11.2 The GSP Algorithm

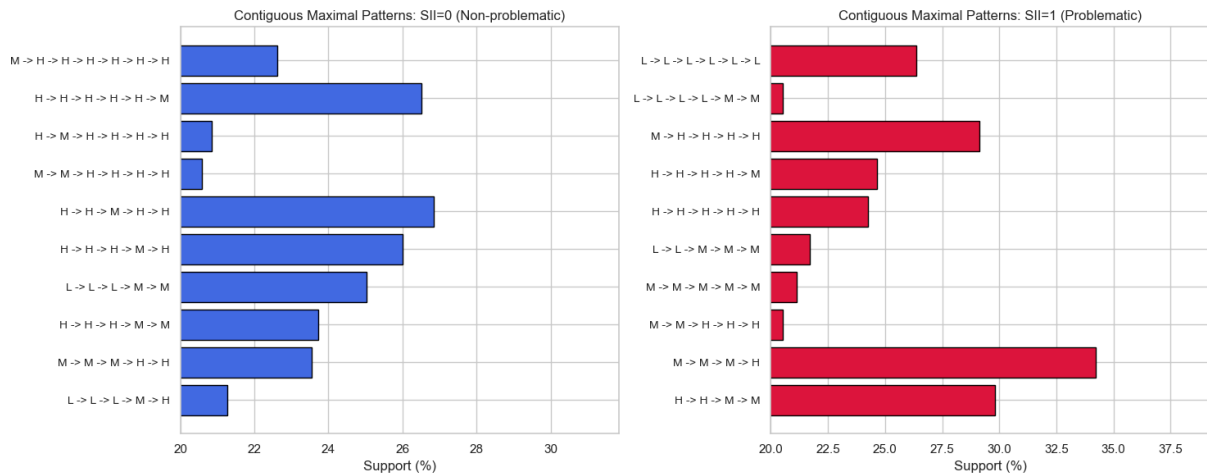
The Generalized Sequential Pattern Mining (GSPM) algorithm expands upon traditional sequential pattern mining by allowing the use of both the elements present within an item sequence and their order to identify sequential relationships among items. Unlike typical basket analysis that considers the presence or absence of each item regardless of its position within the sequence, GSPM allows for the temporal positioning of each element within the sequence to be considered. To this end we have constrained our model with a `max_gap=1` condition which stipulates that there may be no gap between any two sequential elements. For example, a pattern such as $L \rightarrow L \rightarrow L$ does not represent three isolated instances of low activity but rather three separate 5 hour blocks of continuous inactivity, equating to 15 hours. A support threshold was established at 20% and only maximal patterns, i.e., patterns that are included within no other longer frequency pattern, were allowed into the final output.

11.3 Global Frequent Patterns

GSP extracted 70 raw frequent patterns from the full data set that were then reduced to 29 maximal sequential patterns by using subsumption filters. The most frequent of these patterns were $M \rightarrow M \rightarrow L$ (36.4% support; $n = 1291$) and this pattern represented a common progression of activity type from light to rest in the late afternoon/early evening — which is the typical time for the body's "wind down" at night. Some other highly-supported patterns included $H \rightarrow M \rightarrow M \rightarrow M$ (33.0%) as well as $M \rightarrow M \rightarrow M \rightarrow M$ (31.4%). These two latter sequences represent common post-exercise or during exercise moderate level activity sequences respectively. Two of the longer patterns consisted of six segments (or thirty hours). These two long sequences were the $H \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H$ (23.8%) — extended periods of high activity and the $L \rightarrow L \rightarrow L \rightarrow L \rightarrow L \rightarrow L$ (21.8%) — extended periods of low activity/sedentary behavior over an entire day-plus.

11.4 Contrast Pattern Mining

Partitioning by `sii_binary` and mining each cohort independently ($SII=0$: 2,365 subjects; $SII=1$: 1,184) revealed 19 discriminative patterns unique to each group. The most striking contrast: the pattern $L \rightarrow L \rightarrow L \rightarrow L \rightarrow L \rightarrow L$ (30 hours of unbroken sedentary/sleep-level activity) appeared uniquely in the problematic group with 26.4% support — over one quarter of problematic children exhibit this sustained inactivity pattern, while it fails to reach the 20% support threshold in non-problematic children. Conversely, $M \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H$ (35 hours of sustained moderate-to-high activity) was unique to the non-problematic group at 22.6% support.



The previous process of partitioning each part independently (SII=0: 2,365 subjects; SII=1: 1,184) has led to 19 discriminative patterns unique to each group. We noticed a very important difference between the 2 patterns. The pattern $L \rightarrow L \rightarrow L \rightarrow L \rightarrow L \rightarrow L$ (30 hours of unbroken sedentary/sleep-level activity) appeared solely in the problematic group with 26.4% support: this means that over one quarter of problematic children exhibit this sustained inactivity pattern. This same pattern fails to reach 20% support in non-problematic children. Conversely, $M \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H$ (35 hours of sustained moderate-to-high activity) was unique to the non-problematic group at 22.6% support.

The side-by-side horizontal bar charts showcase how non-problematic children's unique patterns (blue, left) are dominated by long chains of H symbols, and so extended periods of vigorous physical activity, while problematic children's unique patterns (red, right) suggest prolonged sedentary stretches. This contrast provides direct, interpretable evidence linking sustained physical inactivity to problematic internet use.

12. Time Series: Conclusion

12.1 What the Models Learned

By analyzing the time series dataset and the extracted sequential behavioral patterns, we were able to identify linkages between physical behavior and digital behavior (sii_binary). Below is a summary of the key insights we have gathered through supervised and unsupervised classification, and sequential pattern mining:

- **Prolonged sedentary behavior as a key discriminator:** Sequential pattern mining via the GSP algorithm revealed a quite contrasting lifestyle pattern between subjects belonging to the two classes. The pattern $L \rightarrow L \rightarrow L \rightarrow L \rightarrow L \rightarrow L$ (representing 30 hours of continuous, unbroken low activity or rest) is highly specific to the problematic internet use group, achieving 26.4% support (affecting over a quarter of these children) while failing to even reach the 20% minimum support threshold in the non-problematic group. On the other hand, the pattern $M \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H \rightarrow H$ (representing 35 hours of sustained moderate-to-high physical activity) was unique to the non-problematic subjects, with 22.6% support. This directly links sustained physical activity to healthy digital habits.
- **Shapelet 10 as key discriminator:** Through shapelet extraction, Shapelet 10 emerged as the most discriminative subsequence. It achieved an Information Gain of 0.003205 with a split threshold of 4.4712. Additionally it provided an actionable rule: if a subject's movement sequence matches Shapelet 10 within a distance of 4.4712, such subject will be classified as Problematic (Class 1).
- **Signal Variability and Stability:** Explainable AI evaluations via Multivariate PULSAR and LORE local explanations have shown that children are classified as "Healthy" when their activity patterns remain consistently above some key movement thresholds. The counterfactual explanations related to an "At Risk" classification involve a reduction in activity variability and specific changes in rate-of-change (derivative) signals, pointing to rigid, sedentary habits.

12.2 Limitations

- **The Class Imbalance Bottleneck:** Due to the extreme imbalanced nature of the training dataset (approximately 66.6% non-problematic vs approximately 33.4% problematic), almost all classifiers failed to appropriately identify the minority class. Although the PULSAR model exhibited the highest precision for Class 1 (61%), its recall for Class 1 was only 29%, which means that over two-thirds of at-risk children were missed by this classifier. Similarly, KNN-Euclidean also missed nearly two-thirds of children that should be classified as at-risk due to this imbalance..
- **Computational complexity impacts ability to perform hyperparameter tuning:** Calculating DTW distances and extracting shapelets was computationally expensive and severely limited our ability to exhaustively tune hyperparameters. While

KNN-Euclidean was successfully tuned using Optuna, other computationally expensive models (KNN-DTW, MiniRocket, and MultiRocket-Hydra) could only be run with fixed or baseline parameters, potentially limiting predictive performance.

- **Loss of Granularity during Discretization:** To enable sequential pattern mining, the continuous ENMO signal had to be coarsely aggregated via PAA from 200 half-hour time steps to 20 five-hour blocks, then binned into terciles. While needed to characterize larger-scale routines (morning-night wind-downs), this very coarse discretization process discards important detailed transition data regarding fine-grained physical transitions that may contain useful predictive information.
- **LORE Local Explanation Failure Modes:** We sometimes encountered failures when generating local rules based upon LORE. When a child's behavior fell into an area of the feature space that was so dense and homogeneous that no local neighborhood could be generated to build contrastive decision trees (as was true for Test Case 10), LORE would fail to produce a set of local rules explaining how the model made its decision for that particular child; therefore, the resulting predictions for those children lacked interpretability.